

Anticipating demand in carbon-aware scheduling across data centres: guarantees and fair carbon attribution

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Abstract

Shifting deferrable jobs to cleaner hours and locations cuts data-centre emissions, but online schedulers act before future grid conditions and jobs are known. We formulate multi-site, deadline-constrained scheduling of bag-of-tasks jobs as a minimum-cost flow problem and propose CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation), which adds “ghost jobs” for expected arrivals to each hourly plan. We prove that uncertainty about demand alone forces any online algorithm, even one with perfect carbon forecasts, to a competitive ratio of $\Omega(\sqrt{\theta})$, where θ is the ratio of the highest to the lowest unit emission, and myopic planning to at least $(1 + \theta)/2$. Given backstop capacity, CARMA never misses a deadline under any predictions, is optimal under exact predictions and degrades linearly with prediction error. Dual prices of the pooled problem yield a carbon attribution among sites that lies in the core. On 2024 regional grid data for Great Britain, CARMA comes within 8.0–9.8% of the clairvoyant optimum, against 15–16% for the best myopic, protection-level or risk-averse baseline and 19–23% for plain model-predictive control, and cuts attributional emissions by 36–68% relative to carbon-agnostic operation. On a real cluster trace with a demand profile learned from different days, its gap is 20.7% against 26.7–36.1%. Anticipating demand is worth more than perfect carbon foresight. Under stylised marginal-emission accounting, achievable reductions shrink to 14% and CARMA’s edge becomes small, so physical savings depend on curtailed low-carbon supply. Origin- and host-based carbon accounting violate coalitional stability in every week, whereas the dual attribution never does in hindsight.

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Keywords: Carbon-aware computing, Data centre scheduling, Competitive analysis, Minimum-cost flow, Cooperative cost allocation, Carbon intensity forecasting

1. Introduction

Data centres are among the fastest-growing electricity consumers. Efficiency gains kept their energy use nearly flat through the 2010s even as computing output grew several-fold [1, 2]. That buffer is now thinning, and information and communication technology accounts for an estimated 2.1–3.9% of global greenhouse-gas emissions [3]. Large machine-learning workloads have drawn particular attention [4, 5, 6]. The carbon intensity of grid electricity varies by a factor of ten or more across regions and hours, driven by the output of wind, solar and dispatchable plant [7]. Operators can therefore cut emissions without new hardware by moving flexible work to cleaner places and times. Such *carbon-aware computing* [8, 9] is an operations problem: capacity is finite, deadlines bind, and both the grid and the job stream are uncertain.

Batch jobs such as model training, analytics, rendering and backups are a natural target, because they tolerate some delay. Previous work has shown the potential of temporal shifting (running later) [10, 11, 12] and of spatial shifting (running elsewhere) [13, 14]. It has also shown their limits. Sukprasert et al. [15] found that practical reductions from spatio-temporal shifting fall well short of idealised bounds. Most methods are evaluated with a scheduler that reacts to the jobs it has already seen, using point forecasts of carbon intensity. Three questions have received less attention. First, how much is lost because an online scheduler does not know which jobs will arrive next, and can any scheduler guarantee good performance in the worst case? Second, does it pay to price the uncertainty of carbon-intensity forecasts into scheduling decisions? Third, when several sites pool their capacity, how should the resulting emissions be attributed so that every participant is better off than on its own?

This paper addresses these questions with a scheduling policy, guarantees under explicitly stated assumptions, and a controlled evaluation on three years of regional grid data for Great Britain. Its building blocks (receding-horizon planning, network flows, conformal calibration and dual cost allocation) are standard; the contribution lies in their combination for this problem, the lower

bound, the conditions under which the guarantees hold and the empirical evidence. It makes five contributions.

1. **Formulation.** We state multi-site, deadline-constrained, width-limited scheduling of deferrable bag-of-tasks jobs, including the emissions of data transfer, as a minimum-cost flow problem (Proposition 1). Offline clairvoyant optima and every online planning step can therefore be solved to optimality, after quantisation, by fast combinatorial min-cost flow algorithms. We also prove that myopic receding-horizon scheduling can be arbitrarily worse than the clairvoyant optimum even with perfect carbon forecasts (Proposition 2).
2. **Algorithm.** We propose CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation). At every hour, CARMA adds *ghost jobs* to its planning network. These represent expected arrivals, learned from the operator’s history, so the planner reserves scarce low-carbon capacity for work that has not yet arrived. The planning problem stays a single min-cost flow and decisions take tens of milliseconds.
3. **Guarantees.** We prove that uncertainty about demand alone forces every online algorithm, even one with perfect carbon forecasts, to a competitive ratio of $\Omega(\sqrt{\theta})$, about $\sqrt{\theta}/2$ for large θ , where θ is the ratio of the highest to the lowest unit emission (Theorem 1). In an idealised setting with a backstop resource, CARMA never misses a deadline whatever its predictions, so it inherits the θ -competitiveness that every on-time policy has. If, in addition, each episode fits within the planning window, CARMA is optimal when its predictions of demand and carbon are exact, and its excess emissions grow at most linearly in the prediction errors (Proposition 3 and Theorem 2). These assumptions do not hold in our weekly experiments, so we verify the theorem separately in episodes that satisfy them. We also show that the dual prices of the pooled scheduling network give a carbon attribution among sites that lies in the core of the pooling game, so no group of sites gains by leaving (Theorem 3).
4. **Forecasting and recalibration pipeline.** As a tool for the scheduling study, we build a reproducible 1–24-hour carbon-intensity forecaster from a public feed: direct gradient-boosted models with horizon-wise normalised conformal quantiles and adaptive recalibration, which keeps empirical coverage close to target under the decarbonisation drift of

2024. Its quantiles let us test risk-adjusted carbon pricing inside the scheduler. We do not claim a new forecasting method.

5. **Evidence.** Across 52 out-of-sample weeks with three workload traces each, three load levels, two operating regimes and a trace-driven workload from the Alibaba 2018 cluster trace, we compare CARMA with seven implementable online baselines, including a revenue-management protection-level policy and a scenario-based stochastic lookahead that uses the same demand information, with clairvoyant and ablated variants, and with the clairvoyant oracle. The results separate the value of anticipating demand from the value of better carbon forecasts, show when each matters, characterise how performance degrades with prediction errors, and show that origin- and host-based carbon accounting violate coalitional stability in every test week, whereas the dual attribution satisfies it exactly for the hindsight benchmark and in over 99% of the checks for CARMA’s realised emissions.

The results carry an operational message. Once spatial shifting is allowed, the dominant loss comes from demand myopia, not from forecast error. Anticipating demand cuts the gap to the clairvoyant optimum by more than half in the synthetic experiments, and by 43% on a real cluster trace whose demand profile is learned from different days. Perfect carbon foresight or risk-averse carbon pricing adds little by comparison. Without migration, temporal shifting alone achieves much smaller reductions, and near saturation it trades emissions against lateness. We also show the limits of the evidence. The headline reductions are attributional. Under a stylised marginal-emission accounting within the single GB market, most of them are reallocations, and physical savings depend on low-carbon supply that would otherwise be curtailed. The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the model, the forecaster, CARMA and the experimental design. Section 4 establishes the theoretical guarantees, with proofs in Appendix A. Section 5 reports results, Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related work

Carbon-aware workload shifting. Early green data-centre schedulers matched deferrable jobs to on-site solar production, for example GreenSlot [16] and GreenHadoop [11]. Other work exploited electricity-price and carbon differences across sites through geographical load balancing [17, 13] and dynamic

106 right-sizing [18]. With public carbon-intensity feeds, attention moved to grid
 107 emissions. Wiesner et al. [10] quantified the reductions from temporal shift-
 108 ing in several regions, including Great Britain, and proposed forecast-based
 109 lowest-window scheduling. Google’s carbon-intelligent compute system sets
 110 next-day virtual capacity curves from carbon forecasts and forecasts of the
 111 fleet’s own demand [8]. It is the closest industrial precedent for using demand
 112 forecasts, but it shapes hourly virtual capacity curves per cluster rather
 113 than planning individual jobs across sites. Cucumber admits delay-tolerant
 114 work only when forecast renewable supply allows it [19], and Caribou shifts
 115 serverless functions between regions [20]. CarbonScaler varies the allocation
 116 of elastic jobs with carbon intensity [12]. Ecovisor exposes the energy system
 117 to applications [21]. Carbon Explorer weighs scheduling against investment
 118 in renewables and storage [22]. Load migration between data centres can
 119 absorb curtailed renewable generation [14], and data-centre demand response
 120 has been surveyed in [23]. Sukprasert et al. [15] bound the potential of spatio-
 121 temporal shifting across 123 regions and find that simple policies capture
 122 most of the achievable benefit. Our study complements theirs by identifying
 123 *which* information limits online schedulers when capacity at clean sites is
 124 scarce.

125 *Online algorithms and guarantees.* Worst-case analysis of carbon-aware
 126 scheduling builds on online search and one-way trading [24], for which
 127 threshold policies achieve optimal competitive ratios. Lechowicz et al.
 128 derived optimal double-threshold algorithms for the online pause-and-resume
 129 problem [25]. With CarbonClipper, they also derived learning-augmented
 130 algorithms with an optimal consistency–robustness trade-off for spatiotem-
 131 poral allocation of a single deadline-constrained workload over a network
 132 of data centres [26]. These works model uncertainty about future carbon
 133 *prices* for one job. We study many jobs that compete for finite low-carbon
 134 capacity, and we show that uncertainty about *demand* creates a distinct
 135 lower bound that carbon forecasts cannot remove. Our guarantees follow
 136 the consistency–robustness paradigm of algorithms with predictions [27].
 137 CARMA’s re-solving of a deterministic plan with expected future demand
 138 is an instance of the certainty-equivalent re-solving heuristics of network
 139 revenue management [28]. Our contribution is not the device itself but
 140 its formulation as a single flow problem for carbon-aware scheduling, the
 141 guarantees and the empirical attribution of value. Table 1 positions the
 142 closest work along the dimensions that matter here.

Table 1: Positioning against the closest work, as modelled in the cited studies (✓: explicitly modelled or provided; -: not modelled).

Work	Many jobs share finite capacity	Information on future demand	Worst-case guarantee	Attribution among sites
Lowest-window temporal shifting [10]	–	–	–	–
Geographical load balancing [13]	✓	–	–	–
Carbon-intelligent compute system [8]	✓	✓	–	–
Online pause-and-resume [25]	–	–	✓	–
CarbonClipper [26]	–	–	✓	–
Flow models of preemptive scheduling [29, 30]	✓ (offline)	–	–	–
This paper	✓	✓	✓ (A1–A2)	✓

143 Our fairness result draws on cooperative game theory: linear production
144 games and flow games have non-empty cores that can be computed from dual
145 prices [31, 32].

146 *Energy-aware scheduling in operations.* Energy-efficient scheduling is a mature
147 topic in manufacturing operations [33], and green scheduling models have ap-
148 peared in this journal, for example for parallel machines with job splitting [34]
149 and for cloud resource allocation [35]. Most such models are deterministic and
150 solved offline, often with metaheuristics. We instead exploit exact network-
151 flow structure and embed it in an online lookahead policy. Network-flow
152 formulations of preemptive scheduling with release dates and deadlines go
153 back to Horn [29] and Federgruen and Groenevelt [30]. Proposition 1 extends
154 this classical construction to multiple sites, job widths, time-varying emission
155 costs and transfer emissions [36, 37]. In the taxonomy of Powell [38], CARMA
156 is a deterministic lookahead policy with a modified (anticipatory) model, in
157 the spirit of model predictive control [39, 40].

158 *Carbon-intensity forecasting.* Forecasts of grid carbon intensity use statistical
159 decomposition [41], machine learning with many exogenous drivers [42], day-
160 ahead regression for building demand response [43], and hierarchical models of
161 the generation mix, as in CarbonCast [44]. For Great Britain, NESO publishes
162 regional forecasts through its Carbon Intensity API [45]. These are average
163 (attributional) intensities. Marginal emission factors, which describe the plant
164 that responds to a change in load, can differ substantially from averages and are
165 the relevant measure for the system-level effect of load shifting [46, 47]. These
166 studies focus on point accuracy. For decisions, calibrated uncertainty matters

as well. Conformal prediction [48, 49, 50] provides distribution-free intervals and has been extended to quantile regression [51], to multi-horizon [52] and dynamic time series [53], and to arbitrary distribution shift through adaptive conformal inference [54]. We use these tools to build a recalibrated forecaster and to test, in a controlled way, whether risk-adjusted carbon costs improve scheduling. The question is motivated by the optimizer’s curse [55] and by the robust-optimisation view that point-estimate solutions can be fragile [56, 57].

3. Material and methods

3.1. Problem setting

We consider an operator that runs a set $\mathcal{S}$ of data-centre sites connected by a wide-area network. Time is divided into hourly slots $t = 0, 1, 2, \dots$. Deferrable batch jobs arrive online. Job j is released at the start of hour a_j at origin site $o_j \in \mathcal{S}$. It carries w_j server-hours of work made of independent tasks (a bag of tasks), can use at most m_j servers concurrently, and must finish before the start of hour $d_j > a_j$. Work may be paused and resumed, and any part of it may run at a site other than its origin. Remote execution requires moving input data. At site s and hour t , $K_s(t)$ servers are free for batch work once the interactive load has been served.

Let $I_s(t)$ be the average carbon intensity of the electricity supplied to site s during hour t (gCO₂/kWh). Executing one server-hour of job j at site s during hour t emits

$$c_{o_js}(t) = e \text{PUE}_s I_s(t) + \mathbf{1}[s \neq o_j] \delta \eta \frac{1}{2} (I_{o_j}(t) + I_s(t)), \quad (1)$$

where e is the IT energy per server-hour, PUE_s is the power usage effectiveness of site s , δ is the data volume moved per migrated server-hour and η is the energy intensity of network transfer (kWh/GB). The transfer term charges network energy at the mean intensity of the two endpoints. With decision variables $x_{js}(t) \geq 0$ (server-hours of job j run at site s in hour t), the offline version of the problem is the linear programme

$$\min_{x, u \geq 0} \quad \sum_j \sum_s \sum_{t=a_j}^{d_j-1} c_{o_js}(t) x_{js}(t) + \sum_j \Pi u_j \quad (2)$$

$$\text{s.t.} \quad \sum_s \sum_{t=a_j}^{d_j-1} x_{js}(t) + u_j = w_j \quad \forall j, \quad (3)$$

$$\sum_s x_{js}(t) \leq m_j \quad \forall j, t \in [a_j, d_j), \quad (4)$$

$$\sum_{j: a_j \leq t < d_j} x_{js}(t) \leq K_s(t) \quad \forall s, t, \quad (5)$$

194 where u_j is work left unfinished at the deadline and Π is a lateness penalty
 195 much larger than any emission cost. When migration is not allowed, $x_{js}(t) = 0$
 196 for $s \neq o_j$; these variables are removed from the problem. With complete
 197 knowledge of arrivals and intensities, the optimum of (2)–(5) is a lower bound
 198 on the emissions of every schedule that serves all jobs by their deadlines, and
 199 we use it as the *clairvoyant oracle*. When emissions are evaluated only for a
 200 subset of jobs (Section 3.5), the oracle still has to serve every job under the
 201 same constraints but is charged emissions only for the evaluated ones, so that
 202 it remains a lower bound on the evaluated emissions of any schedule that
 203 meets all deadlines. Online policies that finish some work late lie outside this
 204 set; Section 3.5 explains how we compare them. An online scheduler has to
 205 fix $x_{.s}(t)$ at the start of hour t . At that point it knows the jobs released so
 206 far, the current intensities $I_s(t)$ and nothing more than forecasts of $I_s(t+h)$
 207 and of future arrivals.

208 **Proposition 1** (Network-flow structure). *Problem (2)–(5) is a minimum-*
 209 *cost flow problem on a network with a source, one node per job, one node*
 210 *per (job, hour) pair, one node per (hour, site) pair and a sink. The source*
 211 *supplies $\sum_j w_j$ units of flow and the sink absorbs them. Arc capacities are w_j*
 212 *(source $\rightarrow j$), m_j ($j \rightarrow (j, t)$) and $K_s(t)$ ($(t, s) \rightarrow \text{sink}$). The only arcs with a*
 213 *cost are $(j, t) \rightarrow (t, s)$, priced at $c_{ojs}(t)$, and a slack arc $j \rightarrow \text{sink}$ priced at Π .*

214 *Proof.* Because the supply equals $\sum_j w_j$ and arc source $\rightarrow j$ has capacity w_j ,
 215 every such arc is saturated, and conservation at the job node is Eq. (3). The
 216 capacity of arc $j \rightarrow (j, t)$ is Eq. (4), conservation at (j, t) splits that hour’s
 217 work across sites, and the capacity of $(t, s) \rightarrow \text{sink}$ is Eq. (5). Conversely,
 218 every feasible flow defines a feasible (x, u) with the same cost. The constraint
 219 matrix of the flow formulation is a node-arc incidence matrix, in which every
 220 column has exactly one $+1$ and one -1 , so it is totally unimodular. $\square$

221 Proposition 1 has a practical consequence. The oracle and every planning
 222 problem solved online (Section 3.3) can be solved by combinatorial min-cost
 223 flow algorithms [36], which return integral optimal flows for integral data. We
 224 quantise work, widths and capacities to 0.01 server-hours (widths rounded up,
 225 capacities down) and costs to 0.01 gCO₂, and solve the quantised network
 226 to optimality with the min-cost flow solver of OR-Tools [58] (version 9.15,
 227 a cost-scaling push-relabel algorithm). The solutions are therefore optimal
 228 for the quantised problem, and the simulator treats remaining work below
 229 0.05 server-hours as complete; it checks every executed allocation against the

230 capacity and width limits up to this tolerance. On the weekly clairvoyant
 231 problems (1,100–2,800 jobs), the flow solver returned the same objective as
 232 the LP solver HiGHS on the unquantised problem to within a relative 5×10^{-5}
 233 and was 3–8 times faster (script `bench_solvers.py`; one CPU core).

234 3.2. Probabilistic carbon-intensity forecasting

235 *Point model.* For each horizon $h = 1, \dots, H$ ($H = 24$ h) we fit a direct
 236 gradient-boosted regression tree model [59] that predicts the change $I_s(t +$
 237 $h) - I_s(t)$ from information available at issue time t . Direct per-horizon
 238 models avoid the error accumulation of recursive strategies [60]. One model
 239 per horizon is pooled across sites, with the site as a categorical feature. The
 240 features are the current intensity; differences between the current value and
 241 lags of 1, 2, 3, 6, 12 and 23 hours; the value observed at the same hour of
 242 the target on the previous day and in the previous week; a 24-hour rolling
 243 mean and mean absolute change (volatility); the national (GB) intensity; the
 244 regional wind, solar and gas shares; and calendar encodings of the target
 245 time (hour of day, day of week, annual sine and cosine). Models are fitted
 246 with absolute-error loss, so they estimate the conditional median. We use
 247 scikit-learn’s histogram gradient boosting [61] with 400 iterations, learning
 248 rate 0.05, 31 leaves and at least 50 samples per leaf. The models use only
 249 information available when the forecast is issued (gaps in the data are filled
 250 causally; Section 3.5), and no weather forecasts. This is deliberate: the goal
 251 is a forecaster any operator can build from a public intensity feed.

252 *Conformal upper quantiles.* A point forecast $\hat{I}_s(t + h | t)$ is turned into cali-
 253 brated upper quantiles by split conformal prediction [48, 49] with normalised
 254 scores. On a calibration window disjoint from training, we compute for each
 255 site s and horizon h

$$r_{s,h}(t) = \frac{I_s(t + h) - \hat{I}_s(t + h | t)}{\sigma_s(t) + \epsilon}, \quad (6)$$

256 where $\sigma_s(t)$ is the 24-hour mean absolute hourly change observed at issue
 257 time and $\epsilon = 5$ gCO₂/kWh. Only pairs whose target $t + h$ also lies in the
 258 calibration window are used, so that no calibration outcome falls in the
 259 tuning period. For a target level β , the upper forecast is $U_s^\beta(t + h | t) =$
 260 $\hat{I}_s(t + h | t) + \hat{q}_{s,h}(\beta_{s,h}(t)) (\sigma_s(t) + \epsilon)$, where $\hat{q}_{s,h}(\beta)$ is the $\lceil (n + 1)\beta \rceil$ -th smallest
 261 of the n calibration scores, the split-conformal quantile. Grid carbon intensity
 262 is not exchangeable over time: 2024 was markedly cleaner than 2022–2023

(Fig. 3). We therefore update the working level online by adaptive conformal inference (ACI) [54]. Once $I_s(t+h)$ is observed,

$$\alpha_{s,h} \leftarrow \alpha_{s,h} + \gamma((1-\beta) - \mathbf{1}[I_s(t+h) > U_s^\beta(t+h | t)]), \quad \beta_{s,h}(t) = 1 - \alpha_{s,h}, \quad (7)$$

with step size $\gamma = 0.005$. At every hour, the outcomes observed in that hour update the levels before the new forecasts are issued, so feedback for horizon h arrives h hours after the forecast. When the working level reaches 1, the bound is set to the largest intensity of the site in the training period, a finite worst case in place of the infinite interval of ACI; when it reaches 0 the bound is zero, and bounds are floored at zero. Gap-filled outcomes give no feedback. For immediate feedback and unbounded intervals, ACI guarantees that the long-run miscoverage frequency converges to $1 - \beta$ for any data sequence [54]. Our variant has delayed feedback, a finite ceiling and a quantile grid of step 0.001, so we do not claim this guarantee and use ACI as an adaptive recalibration whose coverage is checked empirically (Appendix B). The level β then acts as a risk dial that remains roughly calibrated under drift, and with $\beta = 0.5$ the procedure removes level shifts in the point model (a recalibrated median).

3.3. CARMA: carbon-aware anticipatory receding-horizon multi-site allocation

A standard receding-horizon (model predictive control, MPC) scheduler [39] solves, at every hour t , the flow problem of Proposition 1 restricted to pending jobs and the window $[t, t+H)$. Future intensities are replaced by forecasts, and only the first hour of the plan is executed. Such a planner is *myopic about demand*: it sees only jobs that have already arrived. The next result shows that this can be arbitrarily costly even when carbon intensities are known exactly.

Proposition 2 (Myopia can be unboundedly costly). *Even with perfect intensity forecasts, myopic receding-horizon scheduling has no finite competitive ratio with respect to the clairvoyant oracle.*

Proof. Take a clean site A with one server and a dirty site B with ample capacity. Let $c_A(0) = \varepsilon$, $c_A(1) = 0$ and $c_B(0) = c_B(1) = c$, and ignore transfer costs. A flexible job ($w = 1$, $m = 1$) arrives at $t = 0$ with deadline $d = 2$. An urgent job ($w = 1$) arrives at $t = 1$ with $d = 2$. At $t = 0$ the myopic planner sees only the flexible job and defers it to A at $t = 1$ (cost $0 < \varepsilon$). At $t = 1$ both jobs must run in the same hour, one of them at B , so the total cost is c .

296 The oracle runs the flexible job at $t = 0$ and the urgent job at $t = 1$, both at
 297 A , for a total of ε . The ratio c/ε is unbounded as $\varepsilon \rightarrow 0$. $\square$

298 The mechanism in Proposition 2 is common in practice. Low-carbon
 299 capacity is scarce, and a myopic planner spends it on flexible work that could
 300 have waited or run elsewhere, leaving urgent jobs that arrive later to run
 301 on carbon-intensive capacity. CARMA addresses it by anticipating demand
 302 inside the same flow problem, and optionally prices forecast risk.

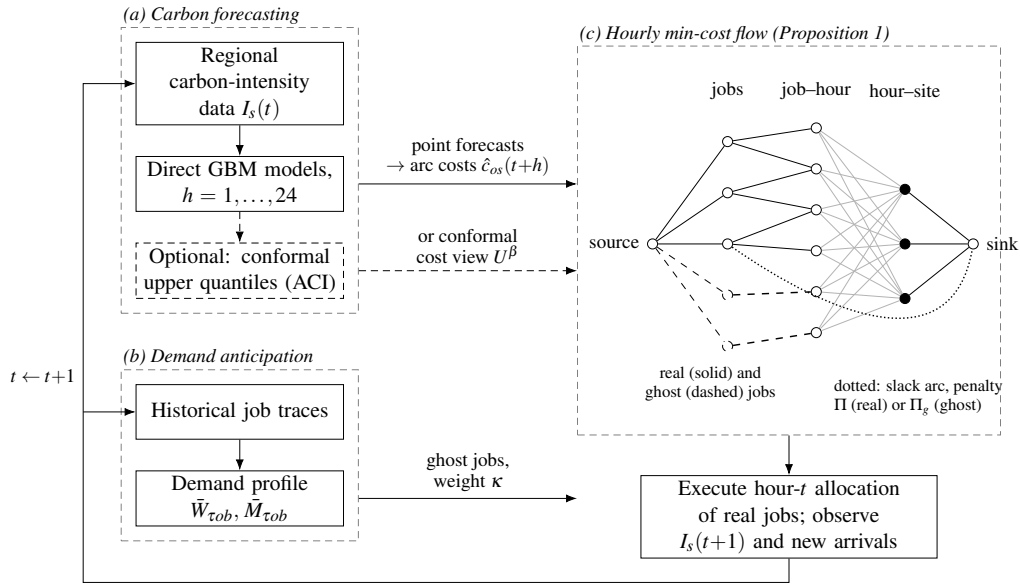


Figure 1: The CARMA decision loop. (a) Carbon-intensity forecasts price the job-hour to hour-site arcs; the selected configuration uses point forecasts, and conormal upper quantiles are an optional cost view. (b) A demand profile learned from past traces generates ghost jobs for expected arrivals. (c) Real jobs (solid) and ghost jobs (dashed) share one min-cost flow network, whose dotted slack arc carries unfinished work to the sink at the lateness penalty. The network is solved exactly every hour, and only the first-hour allocation of real jobs is executed.

303 *Anticipated demand (ghost jobs)*. From the operator's own history we estimate,
 304 for every hour of the week τ , origin o and window class b , the expected work
 305 $\bar{W}_{\tau ob}$ and the expected total width $\bar{M}_{\tau ob}$ of arriving jobs. Window classes are
 306 the bins $\mathcal{B} = \{1, 2, 4, 6, 9, 13, 18, 24\}$ hours. A job is assigned to the largest
 307 bin not exceeding its window, so ghost jobs are never more flexible than the
 308 jobs they represent. Because binning shortens windows, the width of each

ghost job is raised where necessary to at least its work divided by its window, which makes every ghost job window-feasible. At hour t , for each $k = 1, \dots, \ell$ (the look-ahead), CARMA adds to the planning problem a *ghost job* for every (o, b) with work $\kappa \bar{W}_{\tau(t+k)ob}$, width $\kappa \bar{M}_{\tau(t+k)ob}$, release $t + k$ and deadline $\min(t + k + b, t + H)$. Ghost jobs compete with real jobs for capacity, but their slack arc is priced at a moderate penalty Π_g , set to 1.5 times the largest local unit cost in the window, rather than at Π . They can therefore reserve clean capacity when it is economical but never force real work to be late. The weight $\kappa \geq 0$ scales how much of the expected demand is anticipated. With $\kappa = 0$, CARMA reduces to myopic MPC.

Carbon cost view. Future-hour costs in Eq. (1) are evaluated at a forecast of $I_s(t+h)$; for the current hour ($h = 0$) the observed intensity is used. CARMA admits two cost views: the point forecast $\hat{I}$, or the conformal upper quantile U^β of Section 3.2. The second is motivated by the optimizer’s curse [55]: selecting the minimum over noisy estimates is biased towards optimism. The conformal view replaces each future cost by a calibrated quantile, which shifts it by a site- and horizon-specific amount proportional to the local volatility. For $\beta > 0.5$ this penalises volatile sites and distant hours more than stable sites and near hours. Whether this risk premium helps is an empirical question, and we treat the cost view as a hyperparameter.

Algorithm. Fig. 1 summarises the loop. At each hour t , CARMA (i) observes $I_s(t)$, the free capacity and the pending jobs; (ii) builds the cost cube $c_{os}(t+h)$ for $h < H$ from the chosen forecast view; (iii) adds the ghost jobs for hours $t + 1, \dots, t + \ell$; (iv) solves the resulting min-cost flow problem exactly; (v) executes the first-hour allocations of the real jobs only; and (vi) after observing the realised intensities, updates the ACI levels by Eq. (7) when conformal costs are used. Work that is still pending at its deadline receives a one-hour planning window, which makes it urgent in the next plan; it carries the same penalty as other real work, so no strict priority is imposed. The planning network has $O((n_t + \ell |\mathcal{S}| |\mathcal{B}|) H |\mathcal{S}|)$ arcs, where n_t is the number of pending jobs, so each decision takes a few tens of milliseconds (Section 5). The hyperparameters (κ, ℓ) and the cost view are tuned on validation weeks that precede the test period (Section 5.2).

3.4. Baselines

We compare CARMA with the following online policies and the oracle. **ASAP-Local** is carbon-agnostic: it runs work at its origin as soon as possible,

345 earliest deadline first (EDF). **Greedy-Spatial** also runs work immediately,
 346 but at the site that is currently cleanest; this is geographical load balancing
 347 without temporal shifting [13]. **Forecast-Reserve** books, when a job arrives,
 348 the cheapest forecast (site, hour) slots in its window that still have capacity,
 349 and never revises the booking. This is the lowest-window strategy of [10]
 350 extended to multiple sites. **MPC** is the myopic receding-horizon flow scheduler
 351 with point forecasts. **MPC-Protect** adds a protection level in the spirit
 352 of revenue management: planned work may use only a share $1 - r$ of the
 353 capacity of the sites that are cleaner than the fleet median over the window.
 354 In the current hour, the protected share is first offered to jobs due within τ
 355 hours and then, because unused capacity perishes, to all remaining work. The
 356 parameters (r, τ) are tuned on validation data. **MPC+conformal** uses the
 357 conformal upper quantile at the level that was best for MPC on validation
 358 data. **MPC-Perfect** is MPC given the realised intensities, which isolates the
 359 value of forecast accuracy. **CARMA-Perfect** is CARMA given the realised
 360 intensities, and **CARMA+conformal** is CARMA with the best conformal
 361 cost view. **Scenario-MPC** is a demand-aware comparator that uses the
 362 same information as CARMA in a different way. At every hour it draws
 363 K arrival scenarios for the next ℓ hours from the historical job traces from
 364 which CARMA’s profile is estimated (the jobs of one historical week, or trace
 365 day, at the same hours), and solves a two-stage stochastic programme by
 366 sample-average approximation: the first-hour allocation of the pending jobs is
 367 shared, while the later allocation and the scenario jobs are planned separately
 368 in each scenario, and the mean cost is minimised. Unserved scenario work
 369 carries CARMA’s ghost penalty Π_g . The coupling across scenarios breaks the
 370 network structure, so Scenario-MPC is solved as an LP with HiGHS; (K, ℓ)
 371 are tuned on validation data. **Oracle** solves (2)–(5) with full knowledge of
 372 arrivals and intensities.

373 3.5. Data and experimental design

374 *Carbon intensity.* We use half-hourly regional carbon intensity for Great
 375 Britain from the Carbon Intensity API of the National Energy System Op-
 376 erator (NESO) [45] from 1 January 2022 to 31 December 2024. Values are
 377 averaged to hourly resolution. NESO does not meter regional intensity. For
 378 past periods, the regional endpoint returns a single modelled estimate per
 379 half-hour (in a field named `forecast`), and we treat this estimate as the
 380 realised intensity. The oracle and the “Perfect” variants are therefore clair-
 381 voyant with respect to NESO’s published regional values, and all emission

382 figures inherit their modelling error. NESO’s forward regional forecasts are
 383 not archived, so they cannot serve as a historical benchmark for our forecaster.
 384 Four regions host the candidate sites: South Scotland, North West England,
 385 West Midlands and London. They span the north–south gradient of the
 386 British grid, from wind- and nuclear-rich South Scotland to gas-dependent
 387 London, and include the main data-centre markets of London, Manchester
 388 (North West England) and Birmingham (West Midlands). The largest cluster
 389 of the London market, around Slough, lies in NESO’s South England region;
 390 we use the London region as a stylised proxy for a southern, gas-dependent
 391 site. Our first download missed 602 of the 26,304 hours in at least one region,
 392 most of them in two blocks of eleven to twelve days at the start of 2023 and
 393 2024. Re-requesting these days in one-day chunks recovered all but 52 hours,
 394 which the API does not provide: 41 hours on 20–22 October 2023 (in the
 395 tuning period) and 11 hours on 11–12 June 2024. These are filled causally by
 396 carrying the last observation forward, so no filled value uses later data. Filled
 397 hours are excluded from forecast training, calibration, ACI feedback and fore-
 398 cast evaluation. In the scheduling simulation they act as realised intensities;
 399 the two test weeks whose episodes contain them are checked separately in
 400 Section 5.3. The period up to 30 June 2023 is used to train the forecaster,
 401 July–September 2023 to calibrate the conformal scores, October–December
 402 2023 to tune the scheduler, and all 52 weeks starting on Mondays in 2024 as
 403 the out-of-sample test set.

404 *Sites and energy parameters.* Batch capacities are 300, 400, 400 and 600
 405 servers for South Scotland, North West England, West Midlands and London.
 406 This reflects a small, clean northern site and a large site in London; the split
 407 is varied in the sensitivity analysis. Interactive load takes $20\% \pm 15\%$ of each
 408 site’s capacity on a diurnal cycle that peaks in the afternoon. We assume
 409 $e = 0.4$ kWh per server-hour. The assumed PUE values are 1.15, 1.20, 1.25
 410 and 1.30 from north to south, reflecting free-cooling potential. Migrated work
 411 moves $\delta = 2$ GB per server-hour at $\eta = 0.02$ kWh/GB. This value lies within
 412 the wide range of published estimates of transmission energy intensity [62].
 413 The main experiments count only the energy of busy servers, which assumes
 414 that idle servers are switched off or reallocated. The sensitivity analysis varies
 415 η and the capacity split, and it adds an always-on case with an idle draw of
 416 0.2 kWh per server-hour for every batch server.

417 *Workloads.* Job arrivals follow a non-homogeneous Poisson process with a
 418 diurnal peak in the afternoon and 30% lower weekend rates. Jobs originate

419 at the four sites with probabilities 0.10, 0.20, 0.25 and 0.45. Width m_j is
 420 log-uniform on $[4, 64]$ servers and the run time at full width r_j is uniform
 421 on $\{1, \dots, 6\}$ hours, so $w_j = m_j r_j$. A share v of jobs is urgent (deadline
 422 $= a_j + r_j$, no temporal slack); for the others the window is $\lceil f_j r_j \rceil$ hours with
 423 $f_j \sim U[1.5, 4]$, capped at 24 hours. The baseline value is $v = 0.3$. The arrival
 424 rate is scaled so that the offered load equals a fraction $\rho \in \{0.3, 0.5, 0.7\}$ of
 425 the mean free capacity of the fleet. Each test episode covers one week of
 426 arrivals. We also simulate 24 hours of arrivals before and after the week, so
 427 the evaluated week starts and ends under steady-state congestion. Metrics
 428 are computed only for jobs released during the week. For each week and
 429 load level we draw three independent workload traces with fixed seeds, and
 430 every policy runs on the same traces. The demand profile used by CARMA is
 431 estimated from 13 independent historical traces of the same generator. This
 432 favours anticipation, because the profile is then an unbiased estimate of the
 433 arrival rate. We therefore also evaluate a trace-driven workload whose profile
 434 is learned from different days (next paragraph) and profiles learned at the
 435 wrong load (Section 5.5). We study two regimes: *spatio-temporal*, where
 436 migration is allowed, and *temporal-only*, where it is not. All hyperparameters
 437 are tuned separately for each regime.

438 *Trace-derived workload.* To test anticipation on real, non-stationary demand,
 439 we derive a fluid workload from the batch-task table of the Alibaba 2018
 440 cluster trace [63, 64]. The table has 14.3 million task records over eight
 441 days. We keep the 13.3 million terminated tasks with valid timestamps and a
 442 planned CPU request and group them by job (one directed acyclic graph of
 443 tasks), which gives 3.85 million jobs (script `prepare_trace.py`). The trace
 444 records no submission times, so a job’s arrival is the start of its first task. Its
 445 work is the sum over tasks of instances $\times$ run time $\times$ planned CPU request
 446 (core-hours), and its width is the largest planned core count of a single task;
 447 planned requests are not measured utilisation. Both are converted to servers
 448 of 96 cores and scaled so that the offered load in the evaluation days equals
 449 $\rho = 0.5$. Task precedence within a job is ignored. The trace records no
 450 deadlines, so windows are drawn as in the synthetic model: 30% of jobs are
 451 urgent, with a window equal to their run time rounded up to whole hours,
 452 and the others receive 1–23 hours of slack. Origins are assigned at random
 453 with the site probabilities above. To keep the planning problems tractable,
 454 jobs are aggregated into batches by arrival hour, origin and window class,
 455 and the works and widths of a batch are added. Pooling widths relaxes

456 the per-job width limits, because a batch may run its short, wide members
 457 and its long, narrow members at a combined rate that no single member
 458 could sustain. The batched workload is therefore an approximation, and we
 459 test its effect with finer batches that also separate jobs by run-length class
 460 (Section 5.4). CARMA and Scenario-MPC learn from days 1–4 of the trace
 461 and are evaluated on days 5–8, preceded by 24 hours of warm-up. Days
 462 5–8 carry 39% more work than days 1–4, so the learned hour-of-day profile
 463 underestimates evaluation demand by 28%, and hourly work ranges from 0.2
 464 to 3.6 times its mean, so bursts are not anticipated. The four evaluation days
 465 are paired with each of the 52 test weeks of 2024, starting on the Monday,
 466 with three independent draws of origins and deadlines. These 156 episodes
 467 vary the carbon conditions and the synthetic attributes, but they share one
 468 demand chronology.

469 *Metrics and statistics.* The primary metric is operational emissions (tCO₂)
 470 per week under average regional intensity, which is an *attributional* measure
 471 (Section 6 discusses marginal emissions). We report the reduction relative to
 472 ASAP-Local, the gap to the clairvoyant oracle, the share of late work, the
 473 share of work executed away from its origin and the decision time. Reductions
 474 and gaps are means of per-episode ratios, so they cannot be recomputed from
 475 the mean emissions in the tables. Late work is work still outstanding when
 476 its deadline passes; online policies keep processing it, all of it is completed
 477 before the end of the simulation, and its emissions are counted. The gap to
 478 the oracle is an optimality gap only for schedules that meet all deadlines.
 479 Where lateness is negligible, as in the spatio-temporal regime, it measures
 480 the loss of an online policy against the optimum. Where it is not, emissions
 481 and service must be read together, and we also report a service-adjusted
 482 gap. It charges each late server-hour a penalty equal to the largest local
 483 unit cost of the episode, in addition to its emissions, and compares with
 484 an oracle that minimises the same objective: evaluated work may be left
 485 undone at that penalty, while boundary work must still be served on time.
 486 The on-time part of any schedule that serves its boundary work on time is
 487 feasible for this oracle, so its value bounds the adjusted emissions of such
 488 schedules from below. The estimand for comparisons is the mean over the
 489 52 test weeks of the paired difference in gap between a policy and CARMA,
 490 using week-level means over the three traces. Consecutive weeks are serially
 491 correlated (lag-1 autocorrelation of the paired differences ≈ 0.5), so confidence
 492 intervals and two-sided p -values come from a moving-block bootstrap over

493 weeks [65] with blocks of four weeks and 5,000 draws. The p -value is the
 494 share of bootstrap means of the centred series that are at least as far from
 495 zero as the observed mean. We report block lengths of two and eight weeks as
 496 a sensitivity check. p -values are Holm-adjusted jointly over all comparisons
 497 of the main experiment (both regimes and all loads) and separately within
 498 the trace and sensitivity experiments.

499 4. Theoretical guarantees

500 This section establishes a worst-case limit for *every* online scheduler,
 501 prediction-dependent guarantees for CARMA, and a fairness property of the
 502 carbon attribution induced by the pooled scheduling problem. Proofs are
 503 in Appendix A. Throughout, unit costs (Eq. (1), including any backstop
 504 resource introduced below) satisfy $0 < L \leq c_{os}(t) \leq U$, and $\theta = U/L$ is
 505 the *unit-emission-cost ratio*. It refers to unit costs, which include energy
 506 use, PUE, transfer overheads and any backstop, not to grid intensities alone.
 507 Operational costs alone can be zero, because the hourly regional intensity
 508 of South Scotland is 0 gCO₂/kWh in 133 hours of our 2022–2024 data. A
 509 positive floor appears once the embodied emissions of the servers are allocated
 510 per server-hour, which is an accounting convention rather than a claim that
 511 scheduling avoids embodied emissions. As an illustration, a floor of 30 gCO₂e
 512 added to every unit cost, with the largest operational unit cost observed in
 513 2024 (221 g), gives $\theta = 251/30 \approx 8.4$. An online algorithm is Γ -*competitive*
 514 if its emissions never exceed Γ times those of the clairvoyant oracle, for any
 515 arrival sequence and any intensity trajectory.

516 4.1. A universal lower bound under demand uncertainty

517 Competitive analyses of carbon-aware scheduling model uncertainty about
 518 future *prices*, that is, intensities, for a single workload. Examples are online
 519 search and one-way trading [24], the online pause-and-resume problem [25],
 520 and spatiotemporal allocation with deadlines (CarbonClipper) [26]. When
 521 several jobs share scarce low-carbon capacity, difficulty remains even when
 522 all prices are known in advance, because future *demand* is unknown.

523 **Theorem 1** (Lower bound). *Let $\theta > 1$. No online algorithm (deterministic*
 524 *or randomised, with integral or fractional decisions) can be better than $R^*(\theta)$ -*
 525 *competitive, even if it knows all future carbon intensities exactly, where*

$$R^*(\theta) = \max_{1 \leq p \leq \theta} \left\{ 1 + \frac{(p-1)(\theta-p)}{p^2 - p - 1 + \theta} \right\} \geq 1 + \frac{\sqrt{\theta}(\sqrt{\theta}-1)^2}{2\theta - \sqrt{\theta} - 1} = \Omega(\sqrt{\theta}). \quad (8)$$

526 *On the same family of instances, the competitive ratio of myopic receding-*
527 *horizon scheduling (MPC) is at least $(1 + \theta)/2$, a value approached as $p \rightarrow 1$.*

528 For $\theta = 1$ every schedule costs the same and $R^*(1) = 1$; both expressions
529 in (8) tend to 1 as $\theta \downarrow 1$.

530 The bound is established on a two-hour instance and is a quantitative
531 version of Proposition 2, which treats the case $L = 0$. Any algorithm must
532 decide how much flexible work to run now at moderate cost before it learns
533 whether an urgent job will claim the cheap capacity later. The adversary
534 thus raises the effective price of the clean hour from L to U . The resulting
535 order, $\sqrt{\theta}$, is the same as for deterministic online min-search with unknown
536 prices [24]. The difference is where it comes from: here every price is known,
537 and the loss is caused only by the unknown arrival of a competing job. For
538 the illustrative $\theta \approx 8.4$ above, $R^* = 1.81$, whereas myopic MPC can be driven
539 to 4.7. For $\theta = 100$ the values are 5.29 and 50.5. $R^*(\theta)$ is a lower bound
540 derived from one family of instances; we do not know whether any online
541 algorithm attains it. Theorem 1 is a worst-case statement, but it points
542 in the same direction as the average-case evidence of Section 5.3. Carbon
543 forecasts cannot substitute for information about demand, and this motivates
544 the learning-augmented design of CARMA [27].

545 4.2. *Deadline safety, consistency and smoothness of CARMA*

546 We analyse CARMA under two assumptions.

547 (A1) *Backstop capacity.* In every hour, work may also run on a backstop
548 resource of unlimited capacity at a unit cost in $[L, U]$, for example
549 by bursting to a public-cloud region or to reserve servers at a carbon-
550 intensive site. The backstop’s unit cost is known in advance, or its
551 forecast is clipped to $[L, U]$, so the planner never prices it above U . All
552 jobs, real and predicted, are *window-feasible*, $w_j \leq m_j(d_j - a_j)$, and
553 have windows of at most H hours, so every planning problem contains
554 all pending deadlines (this holds by construction in our workloads). The
555 penalties satisfy $\Pi > U$ for real jobs and $\Pi_g \geq U$ for ghost jobs. The
556 implementation enforces window-feasible ghost jobs (Section 3.3) and
557 sets Π_g to 1.5 times the largest local unit cost, which exceeds every unit
558 cost including transfer overheads.

559 (A2) *Episode within the planning window.* The episode $\{0, \dots, T - 1\}$ has
560 length $T \leq H$, as for a daily batch window, so every planning problem
561 extends to the end of the episode.

562 To measure prediction quality at hour t , let F_t be the set of jobs that actually
 563 arrive after t , and $\hat{F}_t$ the ghost jobs used by CARMA. Jobs are compared
 564 after splitting them into *atoms* with proportional widths, which leaves every
 565 planning problem unchanged (Appendix A). Let d_t be the smallest total
 566 work of atoms that must be added to or removed from $\hat{F}_t$ to obtain F_t . Let
 567 $\varepsilon_t = \max_{\tau > t, o, s} |\hat{c}_{os}(\tau | t) - c_{os}(\tau)|$ be the carbon-cost forecast error, and W_t
 568 the total work pending at t plus the larger of the true and predicted future
 569 work.

570 **Proposition 3** (Baseline robustness). *Under (A1), every policy that completes*
 571 *all work by its deadlines is θ -competitive.*

572 **Theorem 2** (Guarantees for CARMA). *Under (A1), for any arrival sequence*
 573 *and any predictions, CARMA completes every job by its deadline (deadline*
 574 *safety) and is therefore θ -competitive. If (A2) also holds, then*

$$\text{ALG} \leq \min \left\{ \theta \cdot \text{OPT}, \text{OPT} + \sum_{t=0}^{T-1} [(U - L) d_t + 2 \varepsilon_t W_t] \right\}. \quad (9)$$

575 *In particular, CARMA is optimal with exact predictions ($d_t = \varepsilon_t = 0$), which*
 576 *is consistency, and its excess emissions grow at most linearly in the prediction*
 577 *errors, which is smoothness.*

578 These properties should be read with care. Robustness by itself does not
 579 distinguish CARMA: by Proposition 3 it holds for any policy that meets
 580 its deadlines. What CARMA adds is deadline safety under *arbitrary* ghost
 581 predictions, together with consistency and smoothness. Deadline safety holds
 582 because ghost jobs carry a softer penalty than real jobs and the backstop can
 583 always absorb real work, so a misprediction can divert real work to dirtier
 584 capacity but never make it late. Without a backstop this is no longer true.
 585 With one server and two hours, a flexible job that could run in the first hour
 586 may be deferred to a cleaner second hour, and an unexpected urgent job in
 587 the second hour then makes one of them late, although an offline schedule
 588 meets both deadlines. A soft ghost penalty does not create capacity. “Exact
 589 predictions” in the theorem means exact predictions of both demand and
 590 carbon; the CARMA-Perfect variant of the experiments has perfect carbon
 591 forecasts but still uses the average demand profile and a finite look-ahead,
 592 so it is not the case $d_t = 0$. The smoothness term charges demand errors
 593 at the spread $U - L$ of unit costs, not at the lateness penalty. The term is

594 informative when predictions are close to the realised demand. For random
 595 arrivals, however, an expected-value profile never matches the realised jobs
 596 exactly. Moreover, each misprediction enters d_t in every epoch in which it is
 597 visible, so the bound in (9) can be loose. In the main experiments, assumption
 598 (A2) is not satisfied (weekly episodes with $H = 24$ h), no backstop is modelled,
 599 and some hours have zero operational cost. The theorem therefore describes
 600 an idealised setting and does not certify the reported numbers; the degradation
 601 profile in the weekly experiments is tested empirically in Section 5.6. Appendix
 602 C checks the theorem numerically in one-day episodes that do satisfy (A1) and
 603 (A2). Robustness θ is also far from the $\Omega(\sqrt{\theta})$ lower bound. Closing this gap
 604 while retaining consistency is left for future work, for example by protecting
 605 deferral decisions with reservation prices [24, 26]. Fig. 2 summarises the
 606 worst-case landscape.

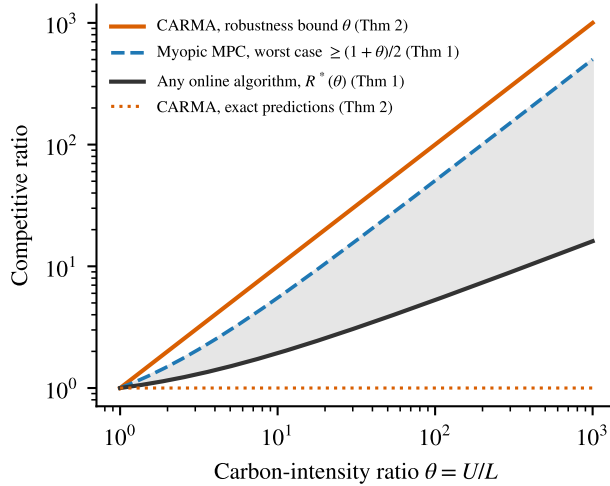


Figure 2: Worst-case landscape as a function of the unit-emission-cost ratio $\theta = U/L$. The curves refer to different information regimes. Without information about future demand, no online algorithm can lie below $R^*(\theta)$, even with exact carbon forecasts (Theorem 1), and myopic MPC can be driven to $(1 + \theta)/2$. Given a backstop (A1), every policy that meets its deadlines, CARMA included, is within θ (Proposition 3). With exact predictions of *both* demand and carbon and an episode within the planning window (A1–A2), CARMA is optimal (Theorem 2).

607 4.3. Fair carbon attribution among sites

608 Pooling capacity reduces total emissions but redistributes them. Work
 609 submitted in London is executed in South Scotland, and Scottish jobs may be
 610 displaced by it. The organisations that operate the sites, or the business units
 611 that submit the jobs, therefore need an attribution of the pooled emissions
 612 that each of them can accept. This is an *internal* allocation problem, like
 613 carbon transfer pricing between units. It is distinct from external GHG
 614 Protocol reporting, where Scope 2 emissions are reported by the entity that
 615 consumes the electricity [66].

616 We model the allocation as a cooperative game. The players are the sites $\mathcal{S}$.
 617 A coalition $C \subseteq \mathcal{S}$ pools the capacity of its sites and serves the jobs submitted
 618 at them. Its stand-alone value $c(C)$ is the optimum of (2)–(5) restricted
 619 to those jobs and sites, which includes the lateness penalty if the coalition
 620 cannot serve its own jobs on time. An attribution $\phi \in \mathbb{R}^{|\mathcal{S}|}$ is *budget-balanced*
 621 if $\sum_o \phi_o = c(\mathcal{S})$. It is in the *core* if $\sum_{o \in C} \phi_o \leq c(C)$ for every coalition C . A
 622 core attribution gives no group of sites a reason to leave the pool and schedule
 623 on its own.

624 Let (λ, ν, μ) be optimal dual variables of the grand-coalition problem: λ_j
 625 for the work constraint (3), $\nu_{jt} \leq 0$ for the width constraint (4) and $\mu_{st} \leq 0$
 626 for the capacity constraint (5). The *dual carbon attribution* is

$$\phi_o = \sum_{j: o_j=o} \left(\lambda_j w_j + \sum_t \nu_{jt} m_j \right) + \sum_t \mu_{ot} K_o(t). \quad (10)$$

627 Site o is charged the carbon shadow price of the work it submits and is
 628 credited ($\mu \leq 0$) with the shadow value of the capacity it contributes. A
 629 site whose clean capacity is worth more than the emissions of its own jobs
 630 therefore receives a net credit, that is, a negative internal charge.

631 **Theorem 3** (Core-stable carbon attribution). *The attribution (10) is budget-*
 632 *balanced and in the core of the pooling game. In particular, it is individually*
 633 *rational: no site is charged more than its stand-alone optimum $c(\{o\})$.*

634 The result follows from Owen’s theorem on linear production games [31];
 635 games defined by network flows are known to be totally balanced [32]. It
 636 concerns the penalised hindsight game defined above. Core stability is one
 637 requirement of an acceptable allocation; it does not imply uniqueness, incentive
 638 compatibility or other notions of fairness, and it does not extend automatically
 639 to the realised emissions of an online policy. The duals are those of the

hindsight problem, which suits after-the-fact accounting. They are available as the optimal node potentials of the min-cost flow; in our experiments we obtain them from an LP solver. Optimal duals need not be unique, so Section 5.7 reports the range of (10) over all optimal duals. It also compares the dual attribution with the Shapley value [67], which is budget-balanced but not guaranteed to lie in the core, and with simpler rules. The simpler rules are *origin-based* (each site is charged the emissions of its own jobs wherever they ran), *host-based* (each site is charged the emissions of the work it executes, the analogue of location-based Scope 2 reporting), and proportional rules.

5. Results

5.1. Carbon-intensity data and forecasts

Fig. 3 shows the regional intensities. Over 2022–2024, mean intensity was 37 gCO₂/kWh in South Scotland, 58 in North West England, 156 in the West Midlands and 163 in London. The regional means indicate large spatial heterogeneity, and it persists hour by hour: the spread between the cleanest and the dirtiest of the four sites has a median of 156 gCO₂/kWh and exceeds 54 g in 90% of hours. This is what makes spatial shifting attractive. Annual mean intensities declined at every site; London’s fell from 202 gCO₂/kWh in 2022 to 125 in 2024. Diurnal profiles peak in the early evening (around 18:00 UTC), and the profile of South Scotland is much flatter than those of the southern sites. The four series are positively correlated (pairwise correlations between 0.51 and 0.87), because national wind output drives all of them.

The forecaster was trained on 1.24 million (issue time, site, horizon) samples in about one minute. In 2024 its mean absolute error (MAE), averaged over horizons and sites, is 31.7 gCO₂/kWh, against 35.3 for persistence and 40.9 for a seasonal-naïve forecast. The advantage is 23–25% at horizons of 1–6 hours, and the difference is significant at all English sites up to 18 hours in Diebold–Mariano tests [68]. Day-ahead skill is limited, because no weather inputs are used. Adaptive recalibration keeps the annual coverage of the upper bounds within 0.003 of target despite the 2024 decarbonisation drift. Appendix B gives the full evaluation (accuracy by horizon and site, coverage, pinball loss).

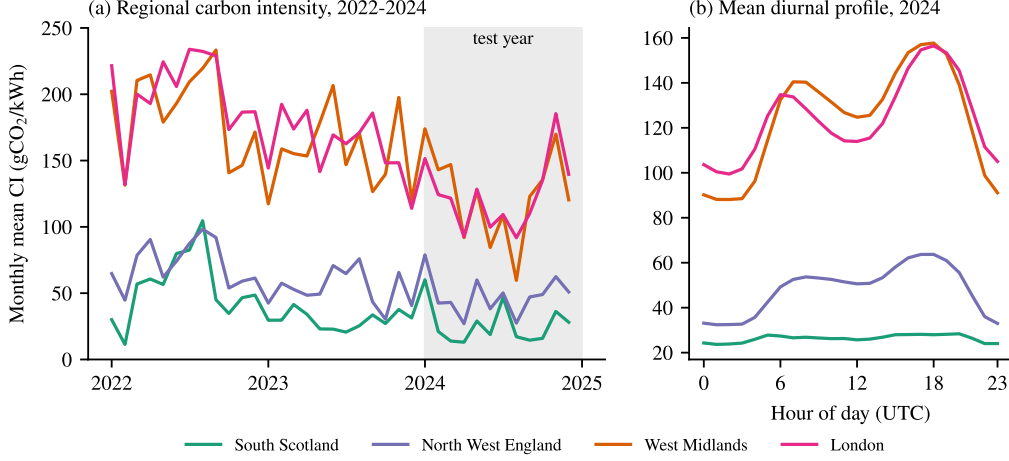


Figure 3: Hourly carbon intensity of the four candidate sites (NESO regional data). (a) Monthly means, 2022–2024; the shaded area is the out-of-sample test year. (b) Mean diurnal profile in 2024 (UTC).

672 5.2. Hyperparameter tuning

673 All policies were tuned on 13 validation weeks (October–December 2023,
674 $\rho \in \{0.3, 0.5, 0.7\}$), separately for each regime. Fig. 4 shows the spatio-
675 temporal regime. Anticipating demand helps across the whole grid: every
676 tested configuration beats myopic MPC, whose mean gap to the oracle
677 is 22.3%. The gap is smallest around $\kappa = 1$, where expected demand is
678 anticipated without inflation: 9.0% for $\ell = 12$ and $\ell = 18$. Over-anticipation
679 ($\kappa = 1.25$ with $\ell = 18$) reserves too much clean capacity and is penalised
680 (12.1%). We selected $\kappa = 1$ and $\ell = 12$ hours, resolving configurations within
681 0.25 percentage points of the best towards the shorter look-ahead. In the
682 temporal-only regime, anticipation brings no benefit. The best configuration
683 ($\kappa = 0.5$, $\ell = 6$; 4.50%) is indistinguishable from MPC (4.50%), and larger
684 κ is slightly worse. For MPC-Protect, the best protection level was $r = 0.3$
685 with $\tau = 1$ h in the spatio-temporal regime (21.0% against 22.3% for MPC).
686 In the temporal-only regime every protection level was worse than MPC (at
687 best 6.0%).

688 The cost view interacts with anticipation (Fig. 4b). For myopic MPC,
689 high conformal levels help. The gap falls from 25.2% at $\beta = 0.5$ to 17.1%
690 at $\beta = 0.9$ and rises again at $\beta = 0.95$ (18.1%), against 22.3% with point
691 forecasts. For CARMA, every conformal level is worse than the point forecast:

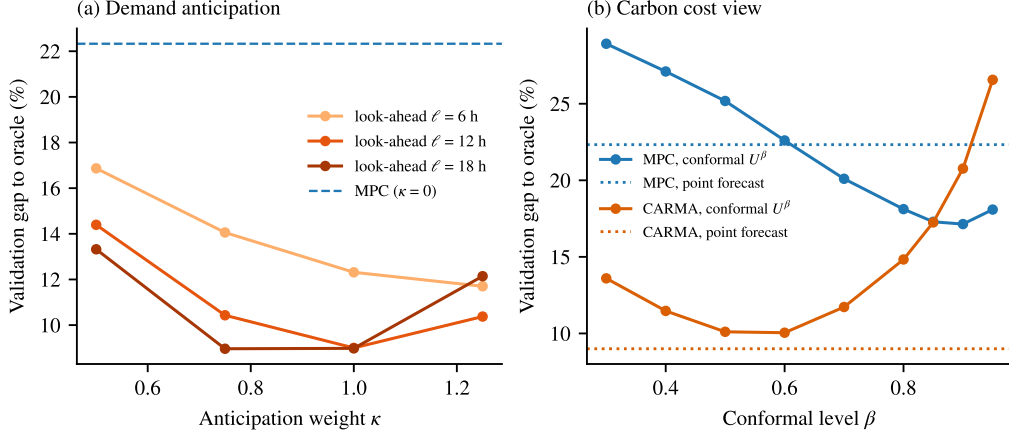


Figure 4: Validation results for the spatio-temporal regime (2023-Q4, mean over 13 weeks and three load levels). (a) Gap to the oracle as a function of the anticipation weight κ and look-ahead ℓ , using point forecasts. (b) Effect of the carbon cost view: conformal quantiles U^β against point forecasts, for myopic MPC and for CARMA ($\kappa = 1$, $\ell = 12$).

the best, $\beta = 0.6$, gives 10.0% against 9.0%. The mechanism is not a uniform risk premium. On the validation weeks, the $\beta = 0.6$ view lies on average only 0.8 g below to 4.9 g above the point forecast at South Scotland, North West England and London, but 17.8 g above it at the volatile West Midlands site. The offset also grows with the horizon, and at $\beta = 0.8$ it rises to 28–128 g at a 24-hour horizon. High levels therefore make distant hours and volatile sites look dirtier, so the planner runs work now and at stable sites, which keeps future clean capacity free. For a myopic planner this is a crude form of anticipation. Once CARMA represents future demand explicitly, the same distortion double-counts it and gives up useful deferrals. The selected CARMA therefore uses point forecasts. For the test we keep MPC+conformal at $\beta = 0.9$ and CARMA+conformal at $\beta = 0.6$ (spatio-temporal) as ablations. In the temporal-only regime, $\beta = 0.5$ was the best conformal view for both.

5.3. Out-of-sample scheduling performance

Tables 2 and 3 report results for the 52 test weeks of 2024, with three independent workload traces per week. Weekly traces contain on average 917, 1,527 and 2,141 jobs (69,400, 115,400 and 162,000 server-hours) at $\rho = 0.3$, 0.5 and 0.7. Fig. 5 shows the gaps to the oracle.

Table 2: Out-of-sample results in the spatio-temporal regime (migration allowed), 52 weeks of 2024×3 traces. Emis.: mean weekly operational emissions (tCO₂, attributional). Red.: reduction relative to ASAP-Local (%). Gap: excess over the clairvoyant oracle (%). Adj.: service-adjusted gap, in which late work is additionally charged at the week’s largest local unit cost. Late: share of work finished after its deadline (%). Mig.: share of work executed away from its origin (%). p : two-sided Wilcoxon signed-rank test of week-mean emissions against CARMA, Holm-adjusted over all comparisons in Tables 2 and 3 of the same regime. The best gap among implementable online policies is in bold.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	Mig.	p
$\rho = 0.3$	ASAP-Local	3.70	0.0	280.1	280.1	0.00	0.0	<0.001
	Greedy-Spatial	1.75	54.1	65.5	65.5	0.00	81.7	<0.001
	Forecast-Reserve	1.38	63.9	26.3	26.3	0.00	83.6	<0.001
	MPC	1.31	65.8	19.2	19.2	0.00	69.4	<0.001
	MPC-Protect	1.40	63.4	28.0	28.0	0.00	80.5	<0.001
	MPC+conformal	1.27	66.8	15.2	15.2	0.00	70.0	<0.001
	MPC-Perfect	1.28	66.6	16.4	16.4	0.00	69.1	<0.001
	CARMA	1.22	68.3	9.3	9.3	0.00	70.1	–
	CARMA+conformal	1.22	68.3	9.5	9.5	0.00	70.4	0.037
	CARMA-Perfect	1.15	70.2	2.4	2.4	0.00	69.9	<0.001
	Oracle	1.13	70.8	0.0	0.0	0.00	70.9	–
$\rho = 0.5$	ASAP-Local	6.08	0.0	147.7	148.2	0.05	0.0	<0.001
	Greedy-Spatial	4.12	33.3	62.2	62.2	0.00	76.4	<0.001
	Forecast-Reserve	3.32	46.9	26.8	26.9	0.02	78.2	<0.001
	MPC	3.22	48.5	22.9	23.0	0.01	50.0	<0.001
	MPC-Protect	3.07	51.1	16.1	16.1	0.00	58.7	<0.001
	MPC+conformal	3.08	51.1	16.4	16.4	0.00	50.6	<0.001
	MPC-Perfect	3.18	49.3	21.1	21.2	0.01	49.4	<0.001
	CARMA	2.92	53.6	9.8	9.8	0.00	50.5	–
	CARMA+conformal	2.94	53.4	10.3	10.3	0.00	50.6	<0.001
	CARMA-Perfect	2.79	56.0	3.8	3.8	0.00	50.2	<0.001
	Oracle	2.70	57.5	0.0	0.0	0.00	51.9	–
$\rho = 0.7$	ASAP-Local	8.29	0.0	71.7	98.5	4.54	0.0	<0.001
	Greedy-Spatial	6.76	19.1	37.9	37.9	0.00	73.3	<0.001
	Forecast-Reserve	6.14	26.9	24.0	30.6	1.12	73.5	<0.001
	MPC	5.94	29.3	20.0	30.9	1.89	31.6	<0.001
	MPC-Protect	5.82	30.9	17.1	21.0	0.69	37.8	<0.001
	MPC+conformal	5.76	31.7	15.8	17.9	0.34	32.1	<0.001
	MPC-Perfect	5.87	30.1	18.6	30.1	1.93	31.4	<0.001
	CARMA	5.41	36.1	8.0	8.1	0.02	31.5	–
	CARMA+conformal	5.44	35.8	8.4	8.5	0.01	31.6	<0.001
	CARMA-Perfect	5.24	38.4	3.9	4.2	0.04	31.0	<0.001
	Oracle	5.07	40.6	0.0	0.0	0.00	32.8	–

Table 3: Out-of-sample results in the temporal-only regime (no migration), 52 weeks of 2024 $\times$ 3 traces. Columns as in Table 2. Greedy-Spatial is omitted because without migration it coincides with ASAP-Local.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	p
$\rho = 0.3$	ASAP-Local	3.70	0.0	21.8	21.8	0.00	<0.001
	Forecast-Reserve	3.27	11.8	7.1	8.6	0.38	<0.001
	MPC	3.23	13.0	5.6	5.6	0.00	0.743
	MPC-Protect	3.30	11.3	7.8	7.8	0.00	<0.001
	MPC+conformal	3.24	12.9	5.7	5.8	0.01	0.232
	MPC-Perfect	3.09	17.4	0.2	0.3	0.01	<0.001
	CARMA	3.23	13.0	5.6	5.6	0.00	–
	CARMA+conformal	3.24	13.0	5.7	5.7	0.01	0.232
	CARMA-Perfect	3.09	17.4	0.2	0.2	0.01	<0.001
	Oracle	3.08	17.6	0.0	0.0	0.00	–
$\rho = 0.5$	ASAP-Local	6.08	0.0	17.6	17.8	0.05	<0.001
	Forecast-Reserve	5.54	9.2	6.6	18.3	3.06	<0.001
	MPC	5.47	10.5	5.1	6.9	0.53	<0.001
	MPC-Protect	5.57	8.8	7.1	8.8	0.50	<0.001
	MPC+conformal	5.50	9.9	5.7	8.4	0.75	<0.001
	MPC-Perfect	5.30	13.7	1.3	4.1	0.76	<0.001
	CARMA	5.46	10.7	4.8	6.1	0.38	–
	CARMA+conformal	5.48	10.2	5.3	7.3	0.54	<0.001
	CARMA-Perfect	5.28	14.0	0.9	2.9	0.54	<0.001
	Oracle	5.24	14.8	0.0	0.0	0.01	–
$\rho = 0.7$	ASAP-Local	8.29	0.0	9.4	22.3	4.54	<0.001
	Forecast-Reserve	7.94	4.3	4.7	30.7	8.15	<0.001
	MPC	7.88	5.0	3.9	42.6	11.95	0.001
	MPC-Protect	8.00	3.6	5.4	43.0	11.62	<0.001
	MPC+conformal	7.92	4.5	4.5	49.6	13.53	<0.001
	MPC-Perfect	7.77	6.5	2.3	49.4	13.83	<0.001
	CARMA	7.87	5.1	3.7	32.6	9.20	–
	CARMA+conformal	7.90	4.7	4.2	39.8	10.87	<0.001
	CARMA-Perfect	7.75	6.8	1.9	41.2	11.63	<0.001
	Oracle	7.62	8.5	0.0	0.0	0.75	–

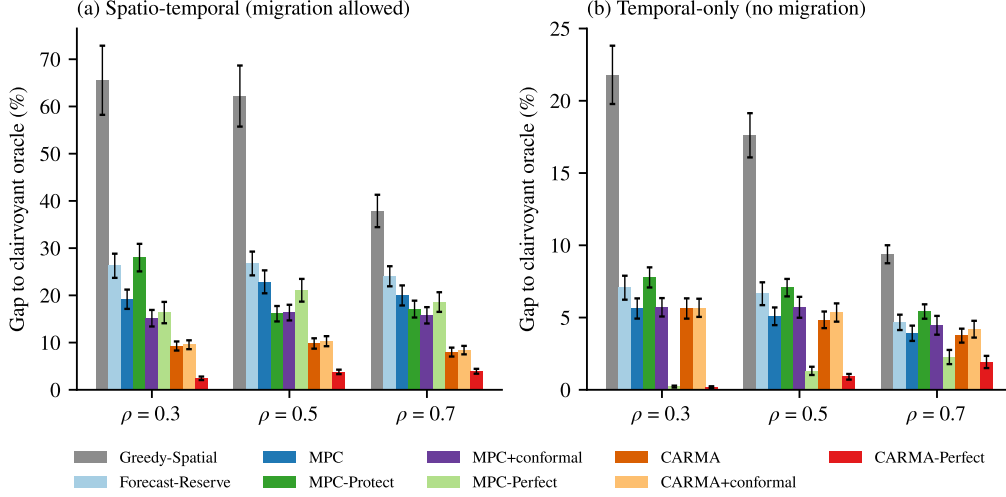


Figure 5: Mean gap to the clairvoyant oracle over the 52 test weeks of 2024 (week means of three traces) with 95% confidence intervals. ASAP-Local is omitted (gap 72–280% with migration, 9–22% without).

710 *Spatio-temporal regime.* With migration allowed, the oracle cuts attributional
711 emissions relative to ASAP-Local by 70.8%, 57.5% and 40.6% at $\rho = 0.3, 0.5$
712 and 0.7. Greedy-Spatial realises only part of this potential (gaps of 38–66%),
713 because it has no temporal flexibility and floods the cleanest site with whatever
714 arrives first. The forecast-driven planners come closer. Forecast-Reserve has
715 gaps of 24–27%, MPC 19–23%, and the two stronger baselines, MPC-Protect
716 and MPC+conformal, 15–28%. MPC-Protect is the best baseline at $\rho = 0.5$
717 (16.1%) and MPC+conformal at the other loads (15.2% and 15.8%). CARMA
718 reduces emissions by 68.3%, 53.6% and 36.1% and has gaps of 9.3%, 9.8%
719 and 8.0%. Relative to MPC this is a further emission reduction of 7.4%,
720 9.3% and 8.8%. Its advantage over the best baseline at each load is 5.9
721 (95% block-bootstrap CI 4.3–8.2), 6.3 (5.2–7.3) and 7.8 (6.7–8.9) percentage
722 points of gap. CARMA emits less than MPC in 51, 51 and 52 of the 52
723 week-means. All comparisons with online baselines are significant after Holm
724 correction ($p < 0.001$). CARMA+conformal is 0.3–0.5 points worse than
725 CARMA ($p \leq 0.04$).

726 CARMA also finishes more work on time under heavy load. At $\rho = 0.7$,
727 0.03% of its work is late, against 1.9% for MPC, 0.7% for MPC-Protect, 1.1%
728 for Forecast-Reserve and 4.5% for ASAP-Local. The service-adjusted gap

729 therefore widens the difference: 8.1% for CARMA against 17.9–30.9% for the
730 forecast-driven baselines. CARMA migrates about the same share of work as
731 MPC (31–70%), whereas MPC-Protect migrates more (38–81%). CARMA’s
732 advantage thus comes from better timing of the use of scarce clean capacity,
733 not from moving more data. For the fleet of 1,700 batch servers at $\rho = 0.5$,
734 CARMA emits 0.30 tCO₂ per week less than MPC, about 16 tCO₂ per year,
735 and 3.16 tCO₂ per week less than carbon-agnostic operation. These absolute
736 figures scale with fleet size.

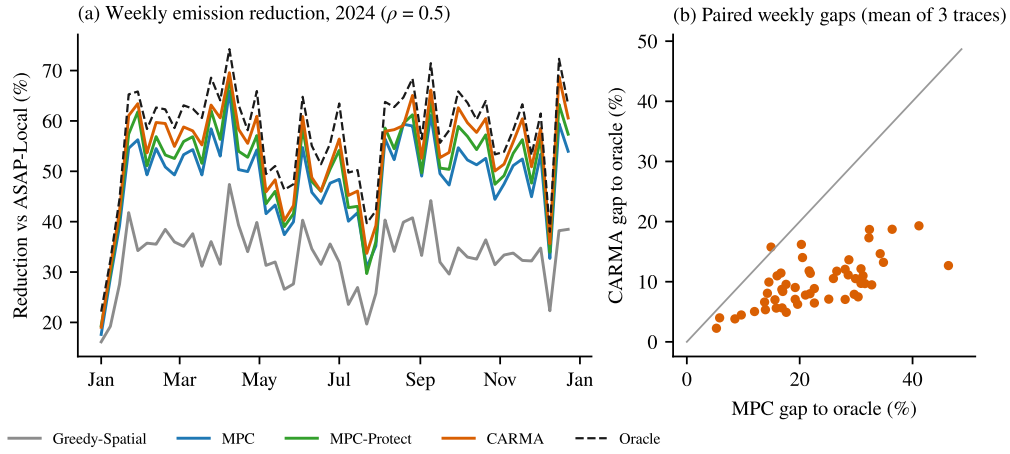


Figure 6: Weekly results in 2024 (spatio-temporal regime, $\rho = 0.5$, means of three traces). (a) Emission reduction relative to ASAP-Local. (b) Paired weekly gaps to the oracle; points below the diagonal are weeks in which CARMA outperforms MPC.

737 *Value of information.* The clairvoyant variants separate two sources of loss.
738 With perfect intensity forecasts, the myopic MPC-Perfect still has gaps of
739 16.4%, 21.1% and 18.6%. Perfect carbon foresight is thus worth only 1.4–
740 2.8 percentage points to a myopic planner, whereas anticipating demand is
741 worth 9.9–13.1 points (MPC versus CARMA). With both, CARMA-Perfect
742 comes within 2.4–3.9% of the oracle. The two kinds of information are
743 complementary. Once demand is anticipated, forecast accuracy is worth 4.1–
744 6.8 points (CARMA versus CARMA-Perfect), about two to four times more
745 than for the myopic planner. The ranking is stable across seasons (Fig. 6a),
746 although the level of achievable reductions varies from week to week with
747 wind conditions.

748 *Temporal-only regime.* Without migration (Table 3), each job can only move
749 in time at its origin. Achievable reductions are much smaller: the oracle saves
750 17.6%, 14.8% and 8.5%. MPC and CARMA are within 3.7–5.6% of the oracle.
751 They are statistically indistinguishable at $\rho = 0.3$ (difference 0.00 points, CI
752 -0.02 to 0.02). CARMA is slightly better at higher loads, by 0.24 (0.15–0.37)
753 and 0.16 (0.08–0.27) points. Both are better than Forecast-Reserve and MPC-
754 Protect. Anticipation brings little benefit here, because each site’s capacity
755 serves only its own jobs, and flexible and urgent work rarely compete for the
756 same clean hours in a way that deferral cannot absorb. The remaining loss is
757 due almost entirely to forecast error: the Perfect variants close the gap to
758 0.2–2.3%. Heavy load exposes a trade-off that the emission metric hides. At
759 $\rho = 0.7$, the capacity at the origin sites is itself insufficient, and every carbon-
760 aware policy delays work: 9.2% of CARMA’s work and 12.0% of MPC’s
761 is late, against 4.5% for ASAP-Local. Under the service-adjusted metric,
762 carbon-agnostic operation (22.3%) is then preferable to MPC (42.6%) and
763 CARMA (32.6%). Without migration and near saturation, temporal shifting
764 should therefore be restricted or combined with explicit service constraints.

765 *Computation.* A CARMA decision takes 9–39 ms on one CPU core, against
766 2–6 ms for MPC; the difference comes from the ghost jobs, which enlarge
767 the planning network. This is negligible for hourly decisions. A clairvoyant
768 weekly oracle with up to about 2,800 jobs is solved exactly in under one
769 second by network simplex.

770 5.4. Trace-driven workload

771 Table 4 reports the Alibaba-trace experiment. Arrivals and job sizes are
772 taken from the trace, and CARMA’s profile is learned from different days,
773 so the profile is biased by 39% and cannot anticipate bursts. The evaluation
774 covers four trace days paired with 52 carbon windows of 2024, with three draws
775 of origins and deadlines each, and 3,072 job batches (65,300 server-hours) per
776 window. Real, bursty demand makes all policies worse than in the synthetic
777 experiment, but the ranking is unchanged. CARMA has a gap of 20.7%.
778 MPC has 36.1%, and MPC-Protect, the best baseline here, 26.7%. Carbon
779 foresight again matters less than demand anticipation: MPC-Perfect reaches
780 only 33.3%, and CARMA’s advantage over MPC-Protect is 6.0 percentage
781 points (95% CI 4.4–8.1; CARMA better in 42 of 52 weeks, $p < 0.001$). The
782 bias of the learned profile costs 2.7 points (CI 1.1–3.8): CARMA with a
783 profile estimated in-sample from the evaluation days reaches 18.0%. With the

784 biased profile but perfect carbon forecasts, the gap would fall by 3.7 points
785 (CARMA-Perfect, 17.0%). The remaining distance to the oracle reflects bursts
786 that no profile anticipates. CARMA also keeps late work low (0.10%, against
787 0.65% for MPC).

Table 4: Trace-driven workload (Alibaba 2018 cluster trace, days 5–8; profile learned from days 1–4), paired with 52 carbon windows of 2024×3 draws of origins and deadlines; spatio-temporal regime, $\rho = 0.5$. Emissions are per four-day window. Columns as in Table 2; the service-adjusted gap is omitted.

Policy	Emis.	Red.	Gap	Late	Mig.	p
ASAP-Local	3.66	0.0	195.3	0.00	0.0	<0.001
Greedy-Spatial	2.50	32.9	92.5	0.00	75.0	<0.001
Forecast-Reserve	1.94	48.6	42.5	0.81	77.4	<0.001
MPC	1.86	50.9	36.1	0.65	52.8	<0.001
MPC-Protect	1.76	53.8	26.7	0.07	64.0	<0.001
MPC-Perfect	1.82	51.9	33.3	0.63	52.3	<0.001
CARMA	1.69	55.7	20.7	0.10	53.2	–
CARMA-Perfect	1.64	57.2	17.0	0.15	52.2	<0.001
Oracle	1.44	62.8	0.0	0.00	56.7	–
CARMA (in-sample profile)	1.65	56.6	18.0	0.01	52.2	<0.001

788 5.5. Sensitivity analysis

789 Table 5 varies one factor at a time around the base case: spatio-temporal
790 regime, $\rho = 0.5$, $v = 0.3$, $\eta = 0.02$ kWh/GB, capacities 300/400/400/600.
791 Each setting uses 26 alternating test weeks of 2024 with two fresh workload
792 traces per week. All policies keep their base-case hyperparameters.

793 *Workload and network.* CARMA has the smallest gap of all online policies in
794 every setting. Except under marginal accounting (discussed below), it emits
795 less than MPC in all 26 weeks of each setting. The benefit of anticipation
796 grows with the share of urgent work, because urgent arrivals are precisely the
797 demand that myopic planners crowd out. With $v = 0.5$, CARMA’s gap is
798 6.7% against 14.9% for MPC-Protect and 21.5% for MPC. With $v = 0.1$, it is
799 11.9% against 16.0% and 23.0%. The ranking is unchanged over a twelve-fold
800 range of transmission energy intensity ($\eta = 0.005$ – 0.06 kWh/GB), and the
801 migrated share of CARMA’s work stays at 49–51%.

Table 5: Sensitivity analysis (spatio-temporal regime; 26 alternating weeks of 2024×2 traces per setting). Oracle red.: emission reduction of the oracle relative to ASAP-Local (%). Gap columns: excess over the oracle (%). Wins: weeks in which CARMA emits less than MPC. Late: share of late work (%), MPC / CARMA. ^aBusy servers draw 0.2 kWh above an idle draw of 0.2 kWh per server-hour that every batch server incurs regardless of the schedule. ^bSchedules are made with average intensities, but emissions are accounted with the stylised marginal emission factor of Section 5.5, and the oracle optimises marginal emissions.

Setting	Oracle red.	Gap to oracle (%)					Wins vs		Late (%)	
		F.-Reserve	Greedy-Sp.	MPC	MPC-Prot.	CARMA	MPC	MPC / CARMA	Late	MPC / CARMA
Urgent share $v = 0.1$	60.8	28.3	77.5	23.0	16.0	11.9	26/26		0.00	0.00
Urgent share $v = 0.5$	55.4	25.4	52.7	21.5	14.9	6.7	26/26		0.09	0.00
Network energy $\eta = 0.005$ kWh/GB	61.2	28.5	68.7	27.0	17.5	10.9	26/26		0.01	0.00
Network energy $\eta = 0.06$ kWh/GB	52.7	28.8	64.1	19.3	14.5	8.4	26/26		0.01	0.00
Load $\rho = 0.4$, profile learned at 0.5	66.0	28.9	74.4	23.2	15.3	12.1	26/26		0.00	0.00
Load $\rho = 0.6$, profile learned at 0.5	50.3	27.5	54.8	23.8	18.7	8.7	26/26		0.31	0.01
Clean-site capacity halved (150)	51.6	25.2	57.1	21.4	16.6	9.5	26/26		0.02	0.00
Clean-site capacity doubled (600)	67.7	32.4	83.0	28.4	16.9	9.2	26/26		0.01	0.00
London capacity doubled (1200)	46.7	27.5	52.0	22.3	18.5	9.5	26/26		0.00	0.00
Always-on servers (idle 0.2 kWh) ^a	18.2	4.7	10.7	3.5	2.5	1.5	26/26		0.01	0.00
Marginal-emission accounting ^b	13.5	10.7	12.0	8.5	9.1	7.9	17/26		0.01	0.00

802 *Capacity layout.* Arrival rates are rescaled with the fleet’s capacity, so the
803 offered load stays at $\rho = 0.5$ of the new capacity. The advantage does not
804 depend on the chosen layout. When the clean South Scotland site is halved
805 (150 servers) or doubled (600), or London’s capacity is doubled, CARMA’s
806 gap stays at 9.2–9.5%. The gaps of MPC (21–28%) and MPC-Protect (16.6–
807 18.5%) stay about two to three times larger. More clean capacity raises
808 the oracle’s reduction (from 51.6% to 67.7%), but a myopic planner still
809 misallocates it.

810 *Misspecified demand profile.* CARMA’s ghost jobs come from a profile learned
811 at $\rho = 0.5$. When the actual load is 20% higher ($\rho = 0.6$), so that CARMA
812 under-anticipates, its gap is 8.7% against 18.7% for MPC-Protect and 23.8%
813 for MPC. When the actual load is 20% lower ($\rho = 0.4$), CARMA over-
814 anticipates and reserves more clean capacity than needed. Its gap rises to
815 12.1%, but it still beats MPC-Protect by 3.2 points (15.3%) and MPC by 11.1
816 points. The trace experiment of Section 5.4 adds a real, larger (39%) bias.
817 Because ghost work carries a lower penalty than real work, the planner never
818 plans real work late in order to serve ghost demand. Without a backstop,
819 however, a misprediction can still delay real work once capacity is exhausted
820 (Section 4.2); the late-work column of Table 5 tracks this.

821 *Always-on servers.* If every batch server draws idle power whether or not it
822 is busy, a large part of the emissions no longer depends on the schedule. The
823 achievable reduction falls to 18.2% for the oracle and 16.9% for CARMA. The
824 ranking is unchanged (gaps of 1.5% for CARMA, 2.5% for MPC-Protect and
825 3.5% for MPC). Carbon-aware scheduling is therefore most valuable when
826 it is combined with right-sizing, so that deferred work actually reduces the
827 number of powered servers [18].

828 *Marginal emissions.* All sites are in one GB market, where gas-fired plant
829 is usually the marginal generator [7, 46]. Moving load between regions then
830 mainly changes *which* region is assigned the average mix, and it abates emis-
831 sions physically only when low-carbon output would otherwise be curtailed.
832 We therefore account emissions with a stylised marginal emission factor (MEF).
833 The MEF is 394 gCO₂/kWh (combined-cycle gas) in every hour except two
834 cases where it is zero. For South Scotland it is zero when Scotland is nearly
835 carbon-free (below 25 g) while GB-wide intensity exceeds 100 g, a proxy for
836 export-constrained hours; this holds in 27% of hours. For all sites it is zero
837 when GB-wide intensity is below 50 g (4% of hours). Under this accounting

the picture changes substantially. The oracle cuts marginal emissions by only 13.5%, and CARMA by 7.3%. The forecast-driven policies are within 7.9–10.7% of the marginal oracle. CARMA is still the best of them, but its advantage over MPC shrinks to 0.6 points (block-bootstrap CI 0.06–0.91), which is not significant after Holm correction ($p = 0.12$; CARMA better in 17 of 26 weeks). The schedules were optimised for average intensity, not for the MEF. The result nonetheless shows that most of the attributional reductions reported above are reallocations within the GB system. The physical benefit depends on whether shifting absorbs otherwise-curtailed low-carbon output, and a scheduler should be driven by a marginal signal when that is the goal.

5.6. Robustness to prediction errors

We test how CARMA degrades as its predictions deteriorate on every fourth test week of 2024 (13 weeks, $\rho = 0.5$, migration allowed, fresh workload traces). Controlled errors are injected into the two kinds of prediction (Fig. 7). These experiments do not satisfy the assumptions of the bound in Theorem 2, so they probe the degradation profile empirically.

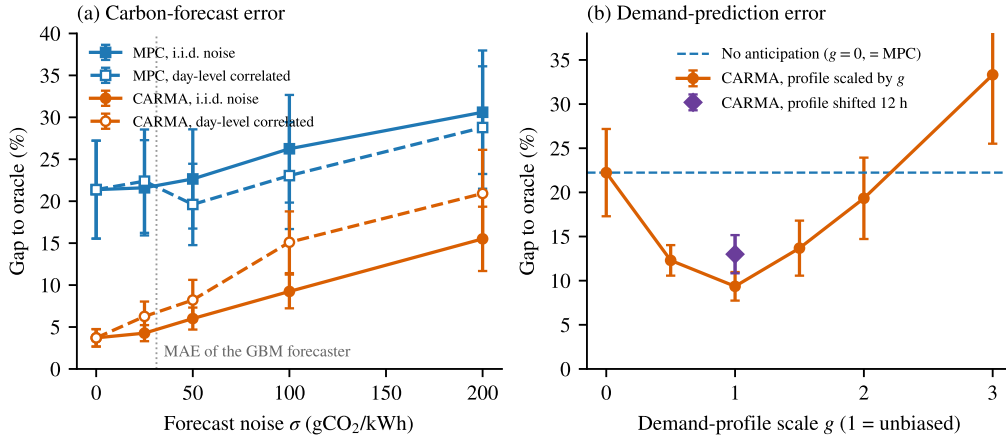


Figure 7: Degradation under controlled prediction errors (13 test weeks of 2024, $\rho = 0.5$, migration allowed; means with 95% confidence intervals). (a) Carbon-intensity forecasts replaced by the realised values plus Gaussian noise of standard deviation σ . The noise is either independent across issue times and horizons (solid) or shared by all forecasts of the same site and target day (dashed, open markers). (b) Demand profile scaled by g , or shifted by 12 hours so that arrivals are predicted at the wrong time of day.

854 *Carbon-forecast error.* With exact forecasts, CARMA’s gap is 3.7% and
855 MPC’s 21.4%. Independent noise is comparatively benign: CARMA’s gap
856 rises smoothly to 4.3%, 6.0%, 9.2% and 15.5% at $\sigma = 25, 50, 100$ and
857 200 gCO₂/kWh. Noise that persists over a whole target day, as when a
858 weather front is missed, is more damaging. The gaps are then 6.3%, 8.2%,
859 15.1% and 20.9%, because the planner cannot average out the error by re-
860 planning. MPC deteriorates from 21.4% to 28.8–30.6% at the largest noise
861 level. At every noise level and type, CARMA remains better than MPC with
862 the same forecasts. Only with the most persistent large errors ($\sigma = 200$) does
863 it fall to the level of MPC with *perfect* forecasts. The actual GBM forecaster
864 (MAE 31 gCO₂/kWh, with errors that are both serially and cross-sectionally
865 correlated) gives a gap of 9.8% in the main experiment. That lies between the
866 independent and correlated cases at comparable error size, so the degradation
867 should not be read as merely i.i.d.

868 *Demand-prediction error.* Scaling the learned demand profile by g produces a
869 V-shaped response with its minimum at the unbiased profile ($g = 1$, gap 9.4%).
870 Under-prediction by half ($g = 0.5$) and over-prediction by half ($g = 1.5$) cost
871 2.9 and 4.3 percentage points. Predicting arrivals at the wrong time of day
872 (a 12-hour shift) costs 3.6 points. Removing anticipation ($g = 0$) returns
873 the MPC level (22.2%). Gross over-prediction ($g = 3$, three times the true
874 demand) is worse than no anticipation (33.3%), because CARMA then holds
875 clean capacity for work that never arrives. A 12-hour shift keeps most of the
876 benefit, which indicates that much of the value comes from anticipating the
877 *volume* of demand rather than its exact timing. Late work was negligible
878 in all 325 runs (at most 0.01% of work). The largest ratio of CARMA’s
879 emissions to the oracle’s over all runs was 1.58.

880 5.7. Fair carbon attribution

881 We evaluate Theorem 3 on the 52 test weeks at $\rho = 0.5$, with the four
882 sites as players. For every week and each of the 15 coalitions we compute
883 the hindsight value $c(C)$, and we also run CARMA with only the coalition’s
884 sites and jobs. Pooling pays: the hindsight optimum of the grand coalition
885 emits 3.37 tCO₂ per week, against 6.70 for the four sites operating separately.
886 With CARMA the figures are 3.63 and 6.88. The question is how to share
887 the reduction.

888 We compare six attributions. The *origin-based* rule charges each site
889 the emissions of its own jobs wherever they run. The *host-based* rule, the

analogue of location-based Scope 2 accounting, charges each site the emissions of the work it hosts. The *work-proportional* rule splits the total in proportion to submitted work, and the *stand-alone proportional* rule in proportion to stand-alone emissions. The *Shapley value* [67] averages marginal contributions over all orders of joining. The *dual* rule is Eq. (10). Because $c(C)$ includes the lateness penalty whenever a coalition cannot serve all of its work, we also evaluate an emissions-only game, in which $c(C)$ is the emissions of the hindsight schedule. Stand-alone coalitions leave work unserved in 94% of weeks, although the amount is small. In that game the dual attribution is rescaled to the emissions total and is no longer covered by the theorem. For CARMA’s realised emissions, the dual attribution is applied to the hindsight value and the online excess over the hindsight optimum is split in proportion to work. A check is violated when a coalition is charged more than it would emit on its own.

Table 6 and Fig. 8 report the results. In the hindsight game the dual attribution satisfies all 728 coalition checks, as Theorem 3 guarantees. The dual prices are not always unique, but over all optimal duals (checked in 9 weeks) the attribution of a site varies by at most 0.011 tCO₂ per week, so the choice of dual does not matter in practice. The four accounting rules violate 22–49% of the checks and fail in every week. The origin-based rule is individually irrational for South Scotland in all 52 weeks and for North West England in 51: their own jobs are displaced to dirtier capacity by imported work, so they are charged more than they would emit on their own. The host-based rule fails for the same two sites in every week, because they are charged for the imported work they host; North West England, for example, is charged 1.50 tCO₂ per week against 0.59 on its own. Scaling stand-alone emissions down is individually rational by construction, but it still leaves 22% of the coalition checks violated. The Shapley value comes close: it is individually rational and violates 3.7% of the checks (44% of weeks), with a largest excess of 0.20 tCO₂. Its site charges are similar to the dual ones. Both credit the clean sites for the capacity they contribute. Under the dual rule South Scotland receives a net credit of 0.91 tCO₂ per week (Shapley 0.77), North West England is close to neutral, and London’s charge (2.81 t) remains well below its stand-alone emissions (3.91 t). Net credits arise in 38% of the (week, site) pairs. The Shapley value also needs all 2^n coalition values, whereas the dual rule comes from a single solve. The emissions-only game gives almost the same picture: the rescaled dual attribution passes every check, and Shapley fails 3.3%.

Table 6: Coalitional stability of carbon attributions (52 test weeks of 2024, $\rho = 0.5$; one 216-hour arrival episode per week). Checks: share of the 52×14 (episode, proper coalition) pairs in which the coalition is charged more than its stand-alone value. Weeks: share of episodes with at least one violation. IR: share of (episode, site) pairs in which a site is charged more than its stand-alone value. Neg.: share of (episode, site) pairs with a net credit. Max. exc.: largest violation (tCO₂). Panel (a) uses the penalised values $c(C)$ of Theorem 3, panel (b) emissions only, panel (c) CARMA’s realised emissions.

Attribution	Checks (%)	Weeks (%)	IR (%)	Neg. (%)	Max. exc.
<i>(a) Hindsight game, penalised cost $c(C)$ (Theorem 3)</i>					
Origin-based	46.4	100.0	49.5	0.0	1.25
Host-based	49.2	100.0	50.0	0.0	4.99
Work-proportional	44.4	100.0	45.7	0.0	1.43
Stand-alone proportional	22.0	100.0	0.0	0.0	0.84
Shapley	3.7	44.2	0.0	40.9	0.20
Dual	0.0	0.0	0.0	37.5	–
<i>(b) Hindsight game, emissions only (dual rescaled; not covered by Theorem 3)</i>					
Shapley	3.3	42.3	0.0	40.9	0.10
Rescaled dual	0.0	0.0	0.0	37.5	–
<i>(c) Realised emissions of CARMA (online game)</i>					
Origin-based	43.1	100.0	44.2	0.0	1.20
Host-based	47.5	100.0	50.0	0.0	4.78
Work-proportional	43.3	100.0	46.2	0.0	1.48
Stand-alone proportional	21.3	100.0	0.0	0.0	0.86
Shapley	3.8	50.0	0.0	39.9	0.13
Adjusted dual (heuristic)	0.8	11.5	0.0	34.6	0.06

928 With CARMA’s realised emissions, which is the practical case, the adjusted
 929 dual attribution remains individually rational for every site in every week. It
 930 violates 6 of the 728 coalition checks (0.8%, in 6 of 52 weeks), all involving
 931 North West England and London, joined in five of them by South Scotland,
 932 and the largest excess is 0.06 tCO₂. Shapley applied to the online values fails
 933 3.8% of the checks, and the four accounting rules fail 21–48%. Stand-alone
 934 coalitions under CARMA leave at most 4.7% of their work late, and unfinished
 935 work emits nothing, so the online checks, if anything, favour secession.

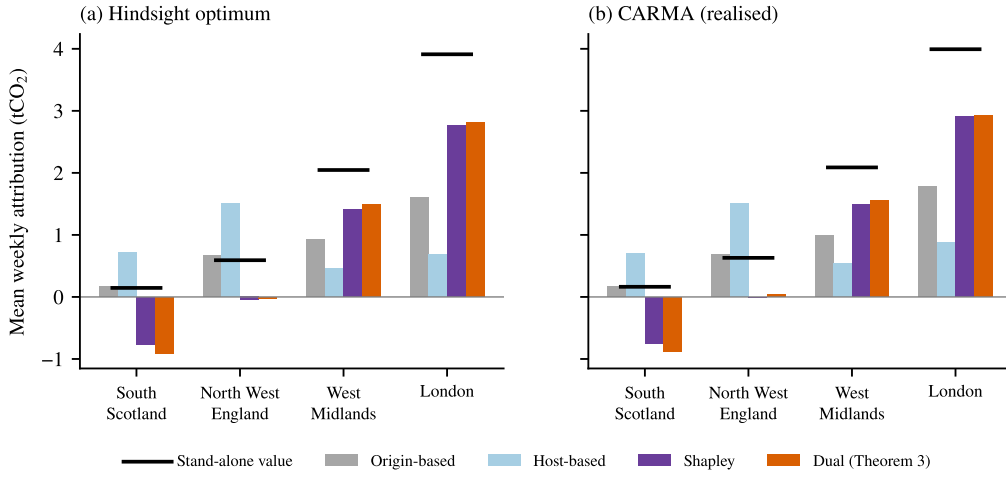


Figure 8: Mean weekly carbon attribution by site under four rules, compared with each site’s stand-alone value (black lines), 52 test weeks of 2024 at $\rho = 0.5$. (a) Hindsight optimum. (b) CARMA’s realised emissions. Negative values are net credits for contributing low-carbon capacity.

936 6. Discussion

937 6.1. Managerial implications

938 *Protect scarce clean capacity.* In multi-site carbon-aware scheduling, the
 939 largest operational loss does not come from imperfect carbon forecasts. It
 940 comes from spending low-carbon capacity on work that could have waited or
 941 run elsewhere. Proposition 2 and Theorem 1 show this in the worst case, and
 942 the experiments confirm it on average. With migration allowed, a myopic
 943 planner with perfect carbon foresight is still 16–21% above the optimum
 944 in the synthetic experiment and 33% above it on the Alibaba trace. Clean

945 capacity at a low-carbon site should be treated as a scarce resource with
 946 an opportunity cost, as in revenue management. A simple protection level
 947 (MPC-Protect) captures part of this value. CARMA captures more, because
 948 its ghost jobs price the protection hour by hour from the operator’s own
 949 history. The result held when the profile was biased by 39% and demand was
 950 bursty.

951 *Invest in forecasts where they pay.* The value of better intensity forecasts
 952 depends on the rest of the policy. For a myopic multi-site planner, perfect
 953 forecasts are worth only 1–3 percentage points; once demand is anticipated
 954 they are worth 4–7 points; without migration they are worth almost the whole
 955 remaining gap. Persistent errors, such as a missed day-ahead wind front,
 956 are more costly than independent ones. Forecasting effort, for example the
 957 integration of numerical weather predictions [44, 43], should therefore target
 958 day-level biases, and in multi-site settings it should follow the fix of demand
 959 myopia.

960 *Use risk premia with care.* The conformal cost views help a myopic planner
 961 but not CARMA. They work by making volatile sites and distant hours look
 962 dirtier, which crudely imitates anticipation. Once demand is anticipated,
 963 the same distortion gives up useful deferrals. Calibrated quantiles are better
 964 suited to reporting and to service-level decisions than to reshaping costs by
 965 default.

966 *Emissions or service near saturation.* Without migration and at high load,
 967 every carbon-aware policy delayed work, and under a service-adjusted metric
 968 carbon-agnostic operation was preferable. Operators should combine temporal
 969 shifting with explicit service constraints, or restrict it, when local capacity is
 970 close to saturation.

971 *Attributional versus physical savings.* The headline reductions are attribu-
 972 tional: they measure how much of the work runs where and when the *average*
 973 regional mix is clean. Within a single market such as Great Britain, moving
 974 load between regions mostly reassigns the average mix. Under a stylised
 975 marginal accounting, the achievable reduction fell from 58% to 14% and
 976 CARMA’s advantage over MPC became small. Carbon-aware shifting is phys-
 977 ically most valuable when it absorbs low-carbon output that would otherwise
 978 be curtailed, for example behind transmission constraints between Scotland
 979 and England, when temporal shifts change which plant is marginal, or when

980 sites lie in different markets. Our marginal accounting is a stylised proxy, not
 981 an estimate of GB-wide abatement, so it indicates the direction and rough
 982 size of this effect rather than its value. In such settings a marginal signal
 983 should drive the scheduler. The scheduling framework is unchanged: only the
 984 cost coefficients of the flow network change. Idle power works in the same
 985 direction. If servers stay powered, deferral saves only their dynamic energy,
 986 and carbon-aware scheduling should be paired with right-sizing [18].

987 *Fair internal carbon accounting.* Pooling halved the hindsight emissions of
 988 the four sites. However, origin-based accounting, which charges each site for
 989 its own jobs wherever they run, would have made South Scotland worse off
 990 than on its own in every week. A pool of organisations or business units
 991 allocated this way is unstable, because its cleanest members have an incentive
 992 to leave. The dual attribution of Theorem 3 charges each site the carbon
 993 shadow price of its demand and credits it for the capacity it contributes. It is
 994 budget-balanced, no coalition would prefer to secede, and it is a by-product
 995 of the scheduling problem. It suits internal carbon pricing and the sharing
 996 of the benefits of carbon-aware scheduling among business units. It is not
 997 a substitute for external GHG Protocol reporting, where negative Scope 2
 998 emissions do not exist [66]. The players must also be entities that could
 999 credibly schedule on their own, such as separate operators, business units
 1000 with their own capacity, or tenants with reserved shares. Because credits
 1001 reward contributed capacity, capacity declarations should be audited.

1002 *Deployment.* CARMA needs a carbon-intensity feed (average or marginal), a
 1003 demand profile computed from the operator’s own job logs and a network-flow
 1004 solver. It decides within tens of milliseconds and returns a plan that is optimal
 1005 for its model, which makes its decisions auditable. Absolute savings scale
 1006 with the fleet: for the 1,700-server fleet simulated here, CARMA saves about
 1007 16 tCO₂ per year relative to myopic control and about 165 tCO₂ relative to
 1008 carbon-agnostic operation, in attributional terms.

1009 6.2. Limitations and future research

1010 The study has several limitations. First, the synthetic workloads use a
 1011 profile learned from the same generator, which favours anticipation. The trace
 1012 experiment partly addresses this concern, but it covers only four evaluation
 1013 days of one cluster, arrival times are first task starts, work is based on planned
 1014 CPU requests, deadlines and origins are synthetic, and jobs are batched. Its

156 episodes vary carbon conditions but share one demand chronology. Second, the regional intensities published by NESO are modelled estimates, used here as realised values, 52 hours of them are missing and filled by persistence, and the marginal analysis relies on a stylised proxy with fixed thresholds, not on a dispatch model or observed curtailment. Third, the facility parameters (PUE, network energy, a 96-core server) are stylised, and embodied emissions are outside the scope except as the floor that makes θ finite. Fourth, the forecaster excludes weather inputs, so its day-ahead skill is limited. Fifth, bag-of-tasks jobs with free preemption and migration are an idealisation: jobs with state, data gravity, bandwidth limits or data-sovereignty constraints would add restrictions that break the pure network structure. Sixth, the guarantees of Theorem 2 assume a backstop resource, a positive emission floor and an episode within the planning window, none of which holds in the simulations. The theorem describes the mechanism and the experiments its empirical profile. Its robustness term is shared by all on-time policies and is far from the $\Omega(\sqrt{\theta})$ lower bound. Seventh, the core property is exact only for the hindsight benchmark.

Several extensions follow naturally. The ghost-demand profile could be replaced by probabilistic arrival forecasts, with the anticipation weight set by a newsvendor-type argument, or by scenario-based planning that treats demand and carbon uncertainty jointly [57]. Tightening robustness towards the lower bound while retaining consistency, for example by protecting deferral decisions with reservation prices [24, 26], is an open theoretical question. The framework can also be driven by marginal emission signals, coupled with right-sizing, on-site renewables and storage [21, 22] and demand-response programmes [23], and evaluated across several electricity markets.

7. Conclusions

This paper studied carbon-aware scheduling of deferrable computing workloads across data-centre sites as a sustainable-operations problem. We showed that the offline problem, including width limits, deadlines and the emissions of data transfer, is a minimum-cost flow, which yields exact oracles and fast online planning. We proved that uncertainty about demand alone forces every online algorithm to a competitive ratio of order $\sqrt{\theta}$, even with perfect carbon forecasts, and that myopic planning does far worse. We proposed CARMA, which adds ghost jobs for expected arrivals to the planning network. Given backstop capacity, it meets every deadline under any predictions, is

optimal under exact predictions and degrades linearly with prediction error. The dual prices of the pooled problem yield a carbon attribution that is budget-balanced and in the core of the pooling game.

In 52 out-of-sample weeks of 2024 on British regional grid data, with migration allowed, CARMA cut attributional emissions relative to carbon-agnostic operation by 36–68%. It came within 8.0–9.8% of the clairvoyant optimum, against 15–16% for the best of the myopic, protection-level and risk-averse baselines, and it did so in almost every week. On a real cluster trace, with a demand profile biased by 39% and bursty arrivals, its gap was 20.7% against 26.7% for the best baseline. Anticipating demand was worth far more than perfect carbon foresight, and correlated day-level forecast errors hurt more than independent ones. Three results qualify these gains. Without migration, reductions are small (9–18%), and near saturation carbon-aware shifting delays work enough that carbon-agnostic operation is preferable under a service-adjusted metric. Under a stylised marginal-emission accounting, the achievable reduction fell to 14% and CARMA’s advantage over MPC was no longer significant, so the physical value of shifting within one grid depends on absorbing otherwise-curtailed low-carbon output. With always-on servers, most emissions no longer depend on the schedule.

Origin-based and host-based carbon accounting violated coalitional stability in every test week. The dual attribution never did in hindsight and failed 6 of 728 checks, by at most 0.06 tCO₂, for CARMA’s realised emissions. Its charges were close to the Shapley value, which it obtains from a single solve. Carbon-aware computing is most effective when schedulers plan for the work that has not yet arrived, not only for the grid conditions that lie ahead, and when its benefits are measured and shared on a sound basis.

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

1085 Funding

1086 This research did not receive any specific grant from funding agencies in
1087 the public, commercial, or not-for-profit sectors.

1088 Data availability

1089 The carbon-intensity data are publicly available from the NESO Carbon
1090 Intensity API [45] under the CC BY 4.0 licence. The data snapshot used in
1091 this study, all code, the workload generator with fixed random seeds, and all
1092 raw experiment outputs are available from the author’s repository (<https://github.com/vinhqdang/Sustainable-Operations-and-Computers>).
1093

1094 Appendix A. Proofs

1095 *Atoms.* Splitting a job j into sub-jobs $j_1, \dots, j_k$ with work w_{j_i} and widths
1096 $m_{j_i} = m_j w_{j_i} / w_j$ does not change the optimal value of any planning problem.
1097 Given a solution for the merged job, the allocation $x_{j_i s}(t) = x_{j s}(t) w_{j_i} / w_j$
1098 satisfies every constraint of the split problem at the same cost. Conversely,
1099 summing the sub-job allocations gives a feasible allocation for the merged
1100 job, because $\sum_i m_{j_i} = m_j$. Splitting also preserves window-feasibility, since
1101 $w_{j_i} / m_{j_i} = w_j / m_j$. We may therefore compare true and predicted job sets
1102 atom by atom.

1103 Appendix A.1. Proof of Theorem 1

1104 Normalise $L = 1$, so $U = \theta$. Site A has one server with cost $p \in [1, \theta]$ in
1105 hour 0 and cost 1 in hour 1. Site B has ample capacity at cost θ in both
1106 hours. Transfer costs are zero. A flexible job ($w = m = 1$, window $\{0, 1\}$) is
1107 released at $t = 0$. The adversary either releases nothing more (scenario 1) or
1108 releases an urgent job ($w = m = 1$, window $\{1\}$) at $t = 1$ (scenario 2). All
1109 costs are known in advance.

1110 Consider any deterministic online algorithm that may split work fraction-
1111 ally, and let $x \in [0, 1]$ be the amount of flexible work it runs at $t = 0$. Running
1112 at B in hour 0 costs $\theta \geq p$ and is dominated, so we may assume this work runs
1113 at A . In scenario 1 the algorithm finishes the remaining $1 - x$ at A in hour 1,
1114 so $\text{ALG}_1 = xp + (1 - x)$, while $\text{OPT}_1 = 1$. In scenario 2, hour 1 must process
1115 $2 - x$ units and A can take only one of them, so $\text{ALG}_2 = xp + 1 + (1 - x)\theta$,
1116 while $\text{OPT}_2 = p + 1$. The ratio $r_1(x) = 1 + x(p - 1)$ increases in x , and the

ratio $r_2(x) = (1 + \theta - x(\theta - p))/(p + 1)$ decreases in x . Hence $\min_x \max\{r_1, r_2\}$ is attained where they are equal, at $x^* = (\theta - p)/(p^2 - p - 1 + \theta) \in [0, 1]$, and its value is

$$1 + \frac{(p - 1)(\theta - p)}{p^2 - p - 1 + \theta}.$$

The adversary chooses p to maximise this value, which gives $R^*(\theta)$. The choice $p = \sqrt{\theta}$ gives the stated closed-form lower bound. For a randomised algorithm, the expected cost in each scenario is affine in x , so it equals the cost of the deterministic fractional algorithm that plays $\mathbb{E}[x]$, and the bound applies against an oblivious adversary. Myopic MPC sees only the flexible job at $t = 0$ and, for any $p > 1$, defers it entirely to the cheaper hour 1 ($x = 0$). Its ratio in scenario 2 is $(1 + \theta)/(p + 1)$, which tends to $(1 + \theta)/2$ as $p \rightarrow 1^+$, so its competitive ratio is at least $(1 + \theta)/2$. $\square$

Appendix A.2. Proof of Proposition 3 and Theorem 2

Proposition 3. A policy that completes all work W on time executes every server-hour at a unit cost of at most U , so its emissions are at most UW . Every server-hour in the oracle's solution costs at least L , and slack costs $\Pi > U \geq L$, so $\text{OPT} \geq LW$. $\square$

Deadline safety. We first show by induction that every pending real job j is window-feasible at the start of every hour: its remaining work satisfies $r_j \leq m_j(d_j - t)$. This holds at release by (A1). Suppose it holds at hour t , and suppose the plan of CARMA leaves some real work $u_j > 0$ unfinished. Because $r_j \leq m_j(d_j - t)$, some hour $\tau \in [t, d_j)$ carries less than m_j units of job j in the plan. Moving a small amount of u_j to the backstop in hour τ keeps the plan feasible, because the backstop has unlimited capacity. It changes the planned cost by a negative amount, because the planner prices the backstop at most at $U < \Pi$ (A1), whatever its forecasts of the other costs are. Since $d_j - t \leq H$ (A1), hour τ lies inside the planning window. This contradicts optimality, so the plan completes all real work. After the first hour is executed, the rest of the plan is a feasible completion, so $r'_j \leq m_j(d_j - t - 1)$ and the induction step holds. No real job is ever late, whatever the ghost jobs are, and θ -competitiveness follows from Proposition 3.

Telescoping. Let σ_t be the state at the start of hour t : the remaining work of all real jobs released by t . Let $V_t(\sigma, G; c)$ be the optimal value of (2)–(5) over hours $t, \dots, T - 1$ for the pending jobs σ and the future jobs G under costs c .

1150 Write $\bar{V}_t = V_t(\sigma_t, F_t; c)$, so that $\bar{V}_0 = \text{OPT}$ and $\bar{V}_T = 0$. For a feasible hour- t
 1151 allocation x with cost $g_t(x)$ and successor state $\sigma^+(x)$, define

$$Q_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), F_t; c), \quad \hat{Q}_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t).$$

1152 By dynamic programming over a deterministic future, $\min_x Q_t(x) = \bar{V}_t$, and
 1153 $Q_t(x_t) = g_t(x_t) + \bar{V}_{t+1}$ for the executed action x_t . Summing over t gives

$$\text{ALG} - \text{OPT} = \sum_{t=0}^{T-1} \delta_t, \quad \delta_t = Q_t(x_t) - \min_x Q_t(x) \geq 0.$$

1154 Ghost jobs are released at $t + 1$ or later, the planning window reaches T
 1155 by (A2), and hour- t costs are observed. The planning problem solved by
 1156 CARMA at hour t is therefore exactly $\min_x \hat{Q}_t(x)$, and x_t is a minimiser.

1157 *One-step loss.* Let $x^* \in \arg \min Q_t$. Since $\hat{Q}_t(x_t) \leq \hat{Q}_t(x^*)$, we have $\delta_t \leq$
 1158 $\Delta(x_t) - \Delta(x^*)$, where $\Delta = Q_t - \hat{Q}_t$. Write $\Delta = D_1 + D_2$ with

$$D_1(x) = V_{t+1}(\sigma^+(x), F_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; c),$$

$$D_2(x) = V_{t+1}(\sigma^+(x), \hat{F}_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t).$$

1159 *Cost error.* An optimal solution under $\hat{c}_t$ is feasible under c , and its cost
 1160 changes by at most ε_t per unit of flow on future-hour arcs. The same holds
 1161 with the roles of c and $\hat{c}_t$ exchanged. Hence $|D_2(x)| \leq \varepsilon_t W_t$ for all x , and
 1162 $D_2(x_t) - D_2(x^*) \leq 2\varepsilon_t W_t$.

1163 *Demand error.* Under (A1), the penalty attached to a window-feasible
 1164 job does not affect any optimal value, provided it is at least U . If a solution
 1165 leaves slack on such a job, the argument used for robustness moves that slack
 1166 to the backstop at no greater cost. True future jobs (penalty Π) and ghost
 1167 jobs (penalty Π_g) can therefore be treated as identical atoms whenever they
 1168 coincide. Let a be a window-feasible atom and G any set of future jobs. For
 1169 every state σ ,

$$Lw_a \leq V(\sigma, G \cup \{a\}) - V(\sigma, G) \leq Uw_a.$$

1170 For the upper bound, add a to an optimal solution for G by running it on the
 1171 backstop at rate $w_a/(d_a - a_a) \leq m_a$ in every hour of its window. For the lower
 1172 bound, delete a 's flow and slack from an optimal solution for $G \cup \{a\}$; each
 1173 deleted unit cost at least $\min\{L, \Pi_a\} = L$, since $\Pi_a \geq U \geq L$. Now transform

1174 $\hat{F}_t$ into F_t by removing $\hat{F}_t \setminus F_t$ and adding $F_t \setminus \hat{F}_t$, one atom at a time. Then
 1175 $D_1(x)$ is a sum of increments, each lying in $[Lw_a, Uw_a]$ (with a minus sign
 1176 for removals) for every x . Therefore $D_1(x_t) - D_1(x^*) \leq (U - L) \sum_a w_a =$
 1177 $(U - L) d_t$.

1178 Combining the two bounds gives $\delta_t \leq (U - L)d_t + 2\varepsilon_t W_t$. Summing over t
 1179 and taking the minimum with the robustness bound proves (9). If $d_t = \varepsilon_t = 0$
 1180 for all t , then $\delta_t = 0$ and $\text{ALG} = \text{OPT}$. $\square$

1181 *Appendix A.3. Proof of Theorem 3*

1182 Write the problem of coalition C as $c(C) = \min\{c^\top z : A_+=z =$
 1183 $b_+(C), A_+z \leq b_+(C), z \geq 0\}$, where $z = (x, u)$. Jobs submitted out-
 1184 side C have zero work and zero width, and sites outside C have zero capacity.
 1185 This forces the corresponding flows to zero, so the problem coincides with
 1186 the stand-alone problem of C . The right-hand side is additive over sites:
 1187 $b(C) = \sum_{o \in C} b^o$, where b^o collects the work and widths of the jobs submitted
 1188 at o and the capacity of site o . The matrix A and the cost c do not depend
 1189 on C . The dual problem $\max\{b(C)^\top y : A^\top y \leq c, y_+ \leq 0\}$ therefore has a
 1190 feasible set that is the same for all coalitions.

1191 Let $y^* = (\lambda, \nu, \mu)$ be an optimal dual solution for $\mathcal{S}$. The slack variables
 1192 make every primal problem feasible, and costs are non-negative, so strong
 1193 duality gives $c(\mathcal{S}) = b(\mathcal{S})^\top y^* = \sum_o (b^o)^\top y^* = \sum_o \phi_o$, which is budget balance.
 1194 For any coalition C , y^* is dual-feasible for C , so weak duality gives $c(C) \geq$
 1195 $b(C)^\top y^* = \sum_{o \in C} \phi_o$, which is the core condition [31]. Expanding $(b^o)^\top y^*$
 1196 gives (10). By Proposition 1 the problem is a min-cost flow, so optimal node
 1197 potentials, together with the reduced costs they induce on saturated capacity
 1198 arcs, also give an optimal dual solution. Any optimal dual solution can be
 1199 used; in our experiments it is returned by the LP solver HiGHS. $\square$

1200 **Appendix B. Forecast evaluation**

1201 Table B.1 and Fig. B.1a summarise forecast accuracy in 2024; hours
 1202 affected by gap filling are excluded. Averaged over horizons and sites, the
 1203 direct GBM forecaster has a mean absolute error (MAE) of 31.7 gCO₂/kWh,
 1204 against 35.3 for persistence and 40.9 for the seasonal-naive forecast. The
 1205 advantage is largest at short and medium horizons: MAE falls by 23% at
 1206 1 hour (6.4 against 8.3) and by 24% at 6 hours (23.6 against 31.2). The
 1207 advantage shrinks with the horizon, and at 24 hours the GBM is worse than
 1208 both benchmarks (42.7 against 40.9). Without weather forecasts, day-ahead

changes in wind output are hard to predict from the intensity history alone [44]. By site, the improvement over persistence is largest in absolute terms for the volatile southern sites (West Midlands 50.2 against 57.7, London 38.0 against 43.4), although their MAE is also the highest. In South Scotland, where intensity is low and often flat, persistence is better overall (14.5 against 16.2). Diebold–Mariano tests [68] with absolute-error loss and HAC variance confirm these patterns. For the three English sites, the GBM is significantly more accurate than persistence at horizons of 1–18 hours ($p \leq 0.002$) and indistinguishable at 24 hours ($p \geq 0.09$). For South Scotland it is significantly better at 1 and 3 hours, indistinguishable at 6 hours and significantly worse from 12 hours onwards.

Table B.1: Forecast accuracy on the 2024 test year (MAE in gCO₂/kWh; mean over sites unless stated).

Model	Horizon h (hours)					All horizons	
	1	3	6	12	24	MAE	RMSE
Persistence	8.3	20.0	31.2	38.6	40.9	35.3	49.2
Seasonal naive (24 h)	40.9	40.9	40.8	40.8	40.9	40.9	57.4
Direct GBM (this study)	6.4	15.1	23.6	34.5	42.7	31.7	41.6

MAE by site over all horizons (persistence / GBM): South Scotland 14.5 / 16.2; North West England 25.5 / 22.3; West Midlands 57.7 / 50.2; London 43.4 / 38.0.

Calibration matters more than point accuracy for the risk-adjusted cost view, and Fig. B.1b shows the effect of distribution shift. Static split-conformal bounds, calibrated on July–September 2023, under-cover in 2024 at every level: at $\beta = 0.9$, annual coverage was 0.868 and monthly coverage ranged from 0.78 to 0.95. With the adaptive recalibration, annual coverage is within 0.003 of the target at all levels (0.499, 0.598, 0.697, 0.797 and 0.898). Monthly coverage stays within about four percentage points of the target (at most 4.2 points), and coverage is uniform across horizons. At $\beta = 0.5$, the recalibrated median lies on average 10.3 gCO₂/kWh below the point forecast, which reflects the downward drift in intensity that the static model does not capture. Coverage is not the whole story. In pinball loss, the adaptive bounds improve on static calibration at levels 0.5–0.7 (15.01 against 15.16 at 0.5) but are worse at 0.8 and 0.9 (9.07 against 8.36 at 0.9). Delayed feedback explains much of this. The level for horizon h is updated only when the outcome is observed, h hours after issue, so a day of under-forecasts drives the level of long horizons

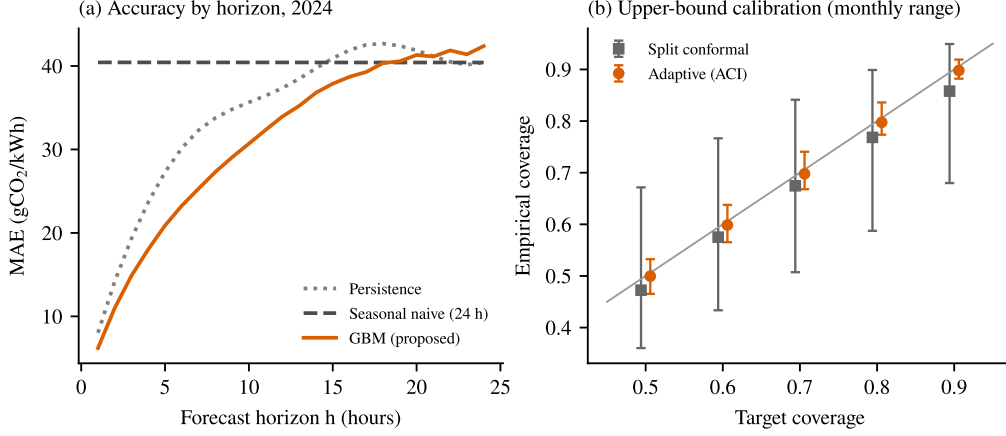


Figure B.1: Forecast evaluation in 2024. (a) MAE by horizon, averaged over sites. (b) Empirical coverage of one-sided upper bounds U^β against the target level β for static split conformal calibration and adaptive conformal inference (ACI). Markers show the annual coverage and bars the range across the 12 months.

towards 1 before any correction arrives. At $\beta = 0.9$, 5.6% of the 2024 bounds (0.5% at $\beta = 0.8$) were at the fallback ceiling, the training-period maximum of the site. These features, together with the finite ceiling and the quantile grid, are why we treat the procedure as an empirical recalibration and do not invoke the coverage guarantee of ACI [54]. Annual coverage says nothing about conditional coverage in the site-hours that the optimiser selects.

Appendix C. Numerical check of Theorem 2

(Results pending.)

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